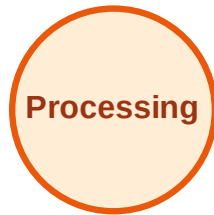


BFS Traversal

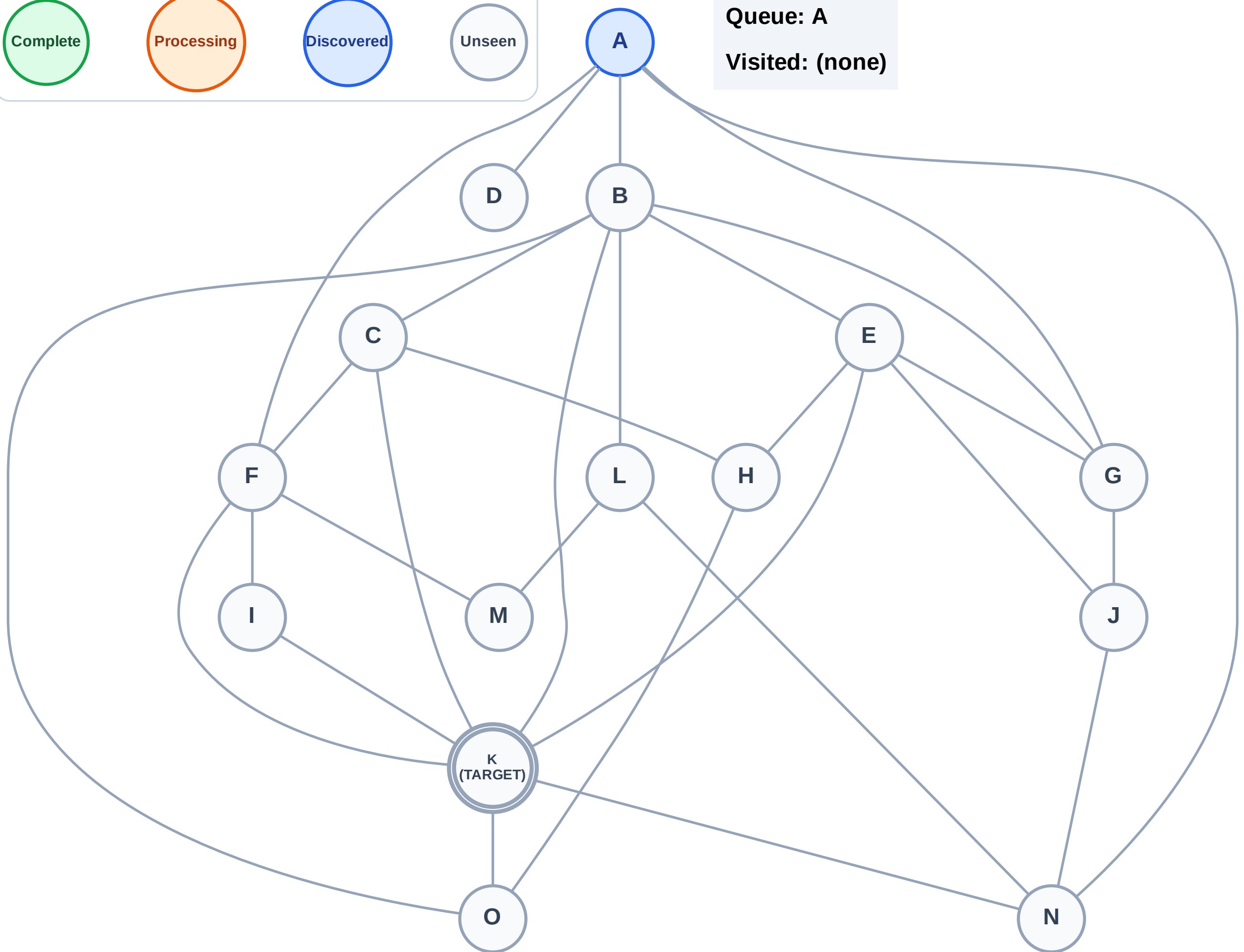
Step 1: Start at A (Seeking target node K)

Legend



Queue: A

Visited: (none)



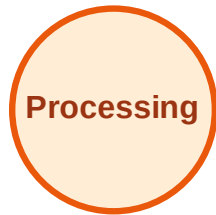
BFS Traversal

Step 2: Process node A

Legend



Complete



Processing



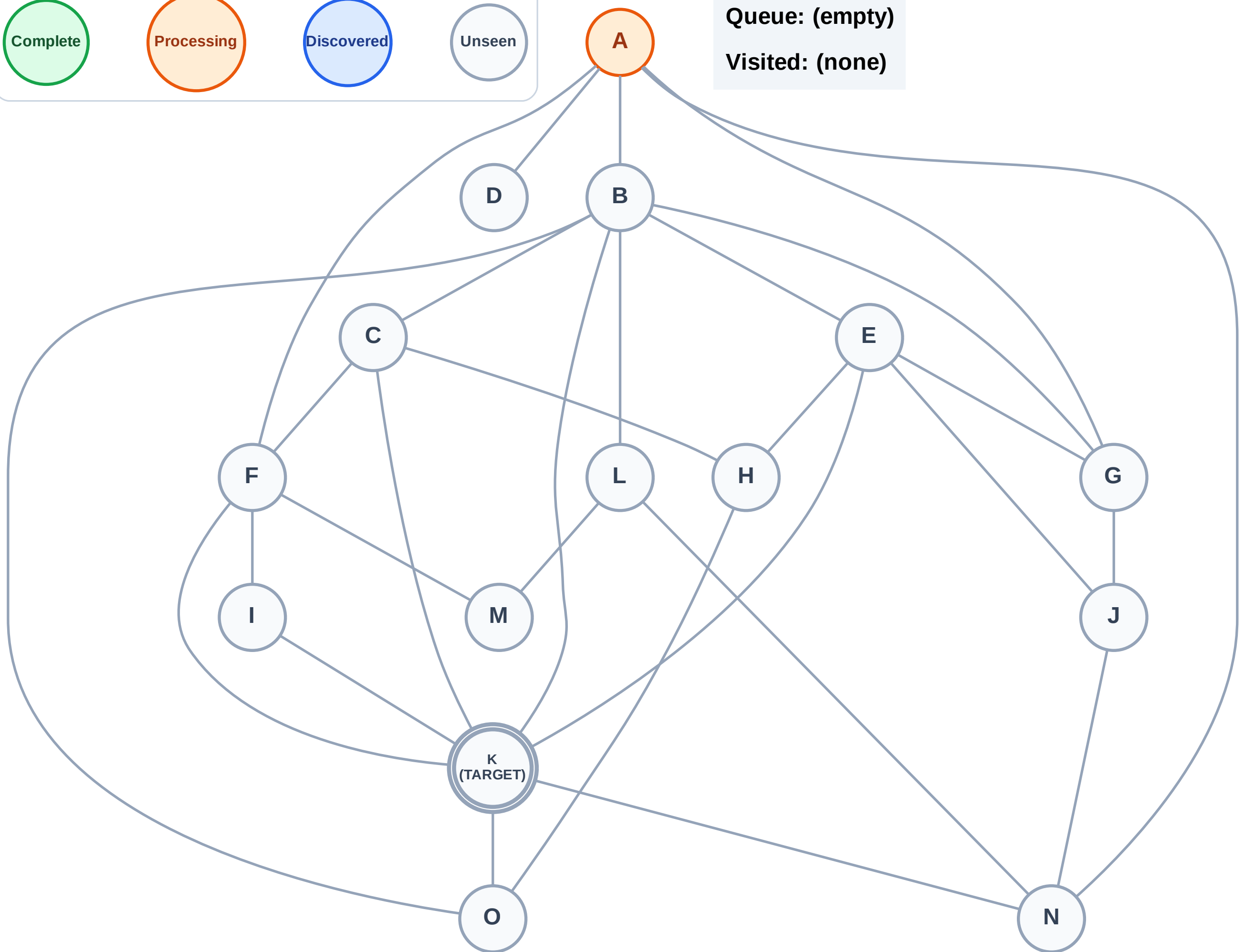
Discovered



Unseen

Queue: (empty)

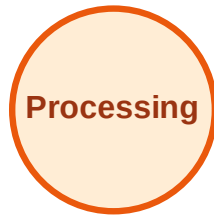
Visited: (none)



BFS Traversal

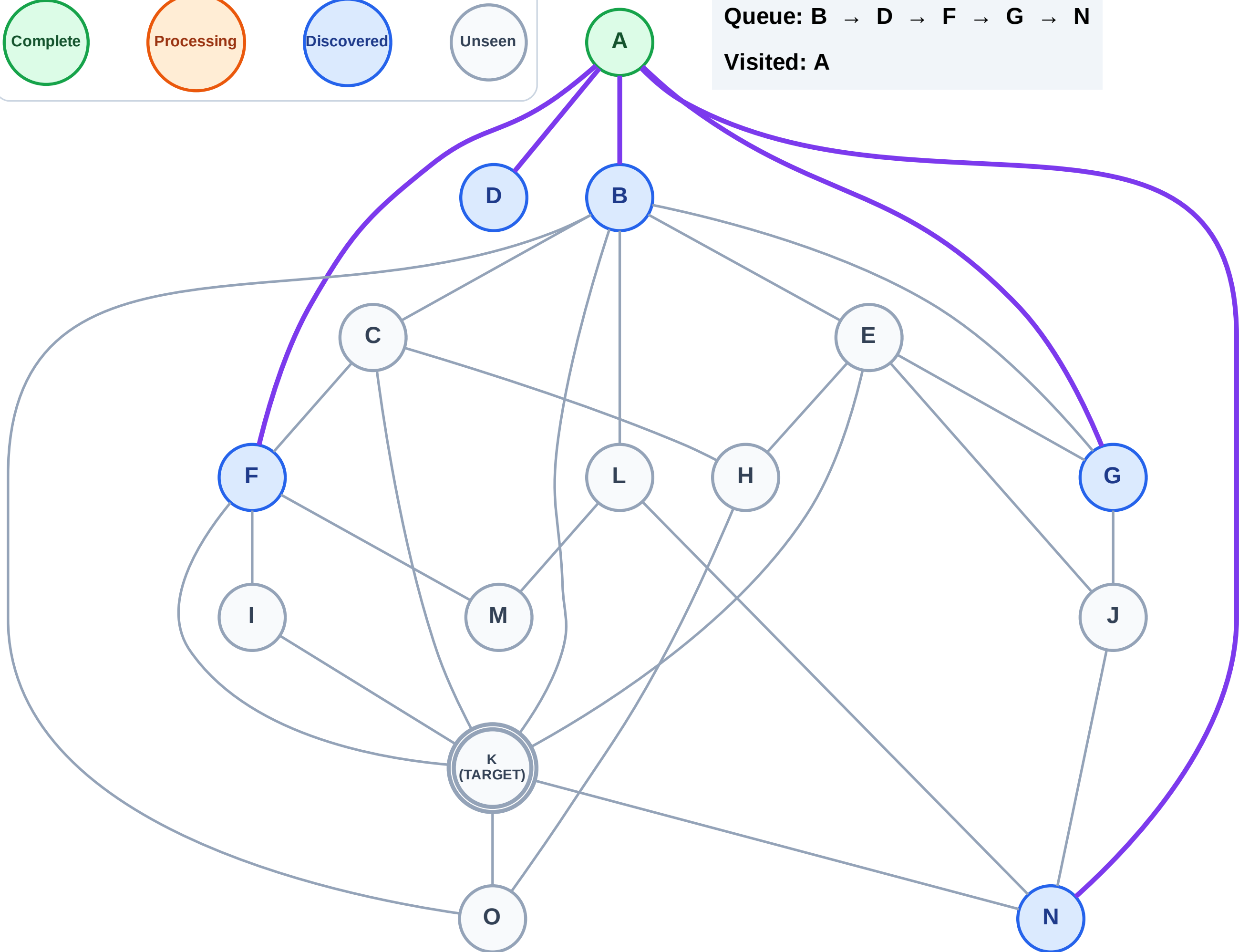
Step 3: Discover B, D, F, G, N from A

Legend



Queue: B → D → F → G → N

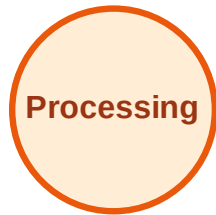
Visited: A



BFS Traversal

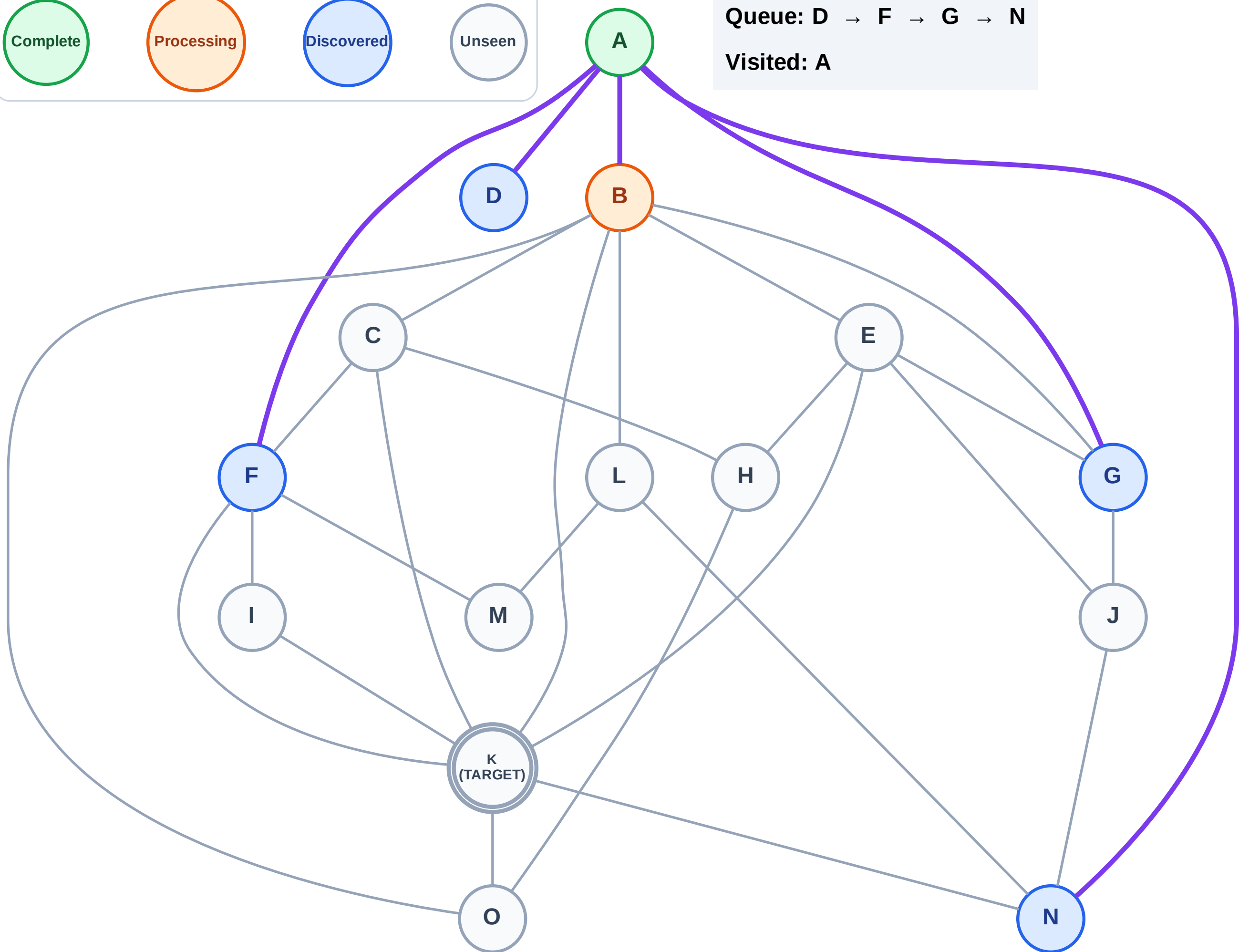
Step 4: Process node B

Legend



Queue: D → F → G → N

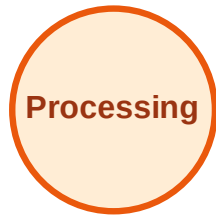
Visited: A



BFS Traversal

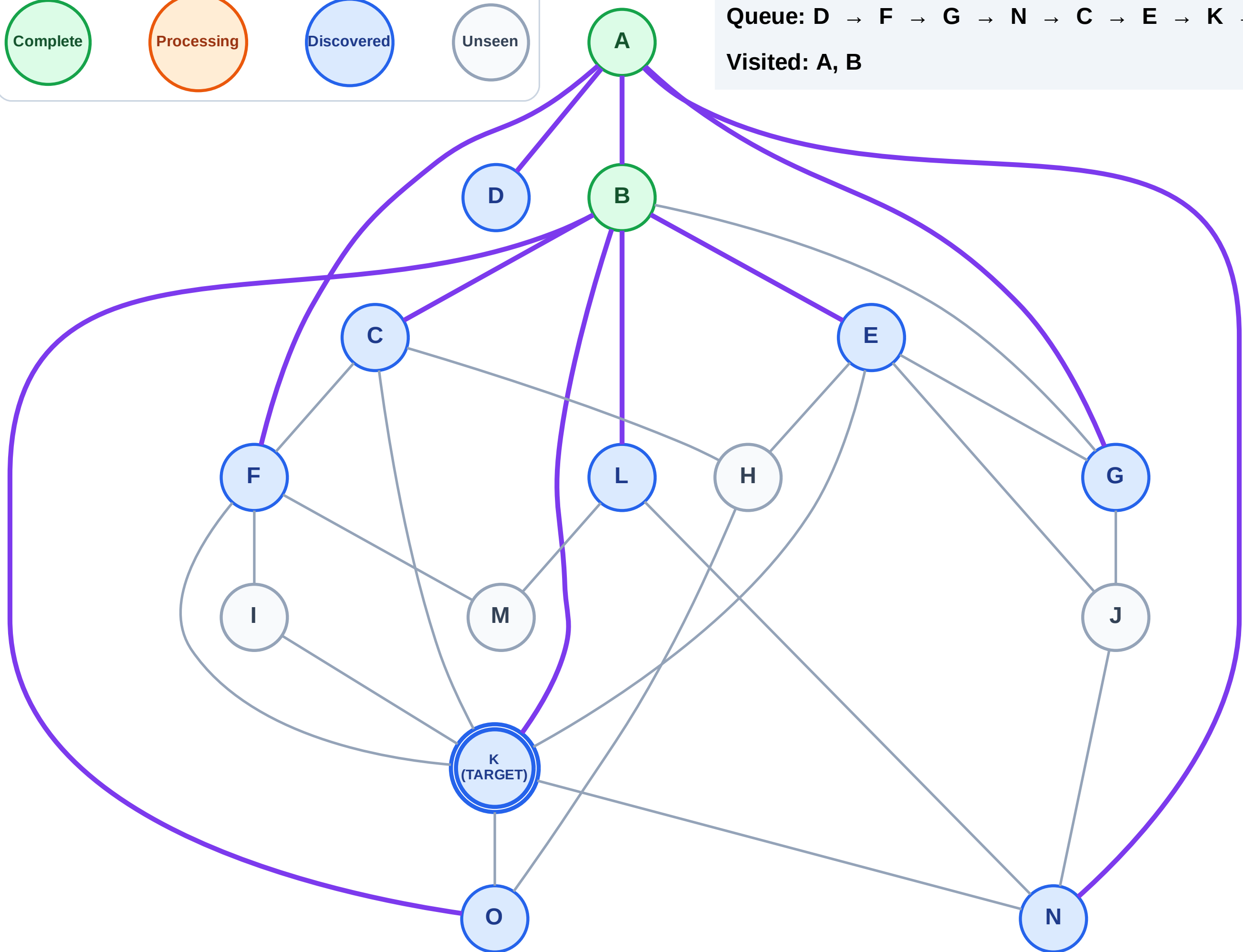
Step 5: Discover C, E, K, L, O from B

Legend



Queue: D → F → G → N → C → E → K → L → O

Visited: A, B



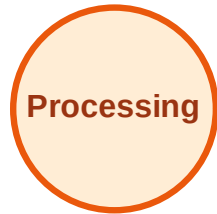
BFS Traversal

Step 6: Process node D

Legend



Complete



Processing



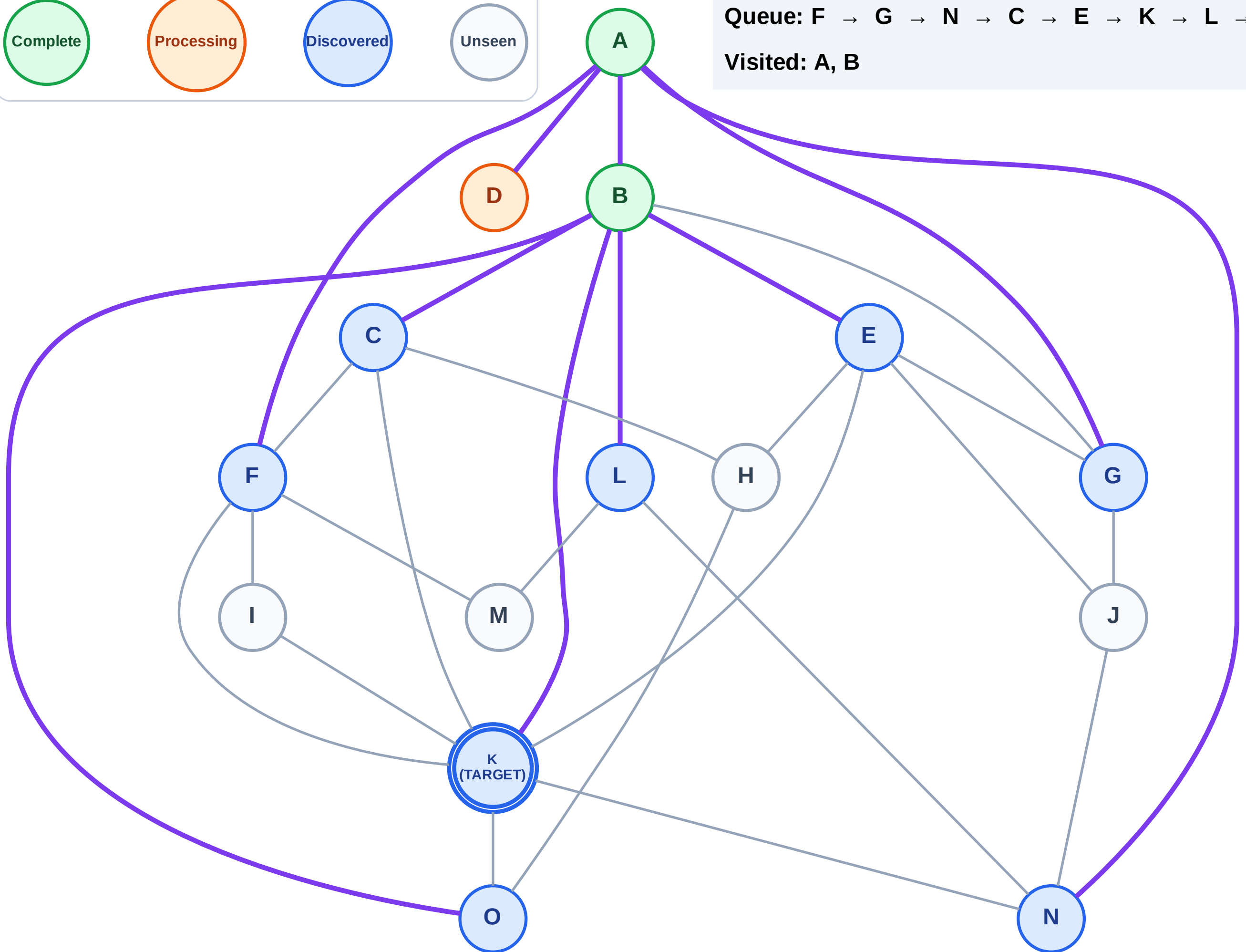
Discovered



Unseen

Queue: F → G → N → C → E → K → L → O

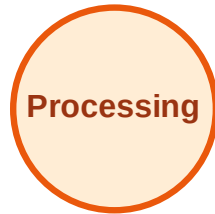
Visited: A, B



BFS Traversal

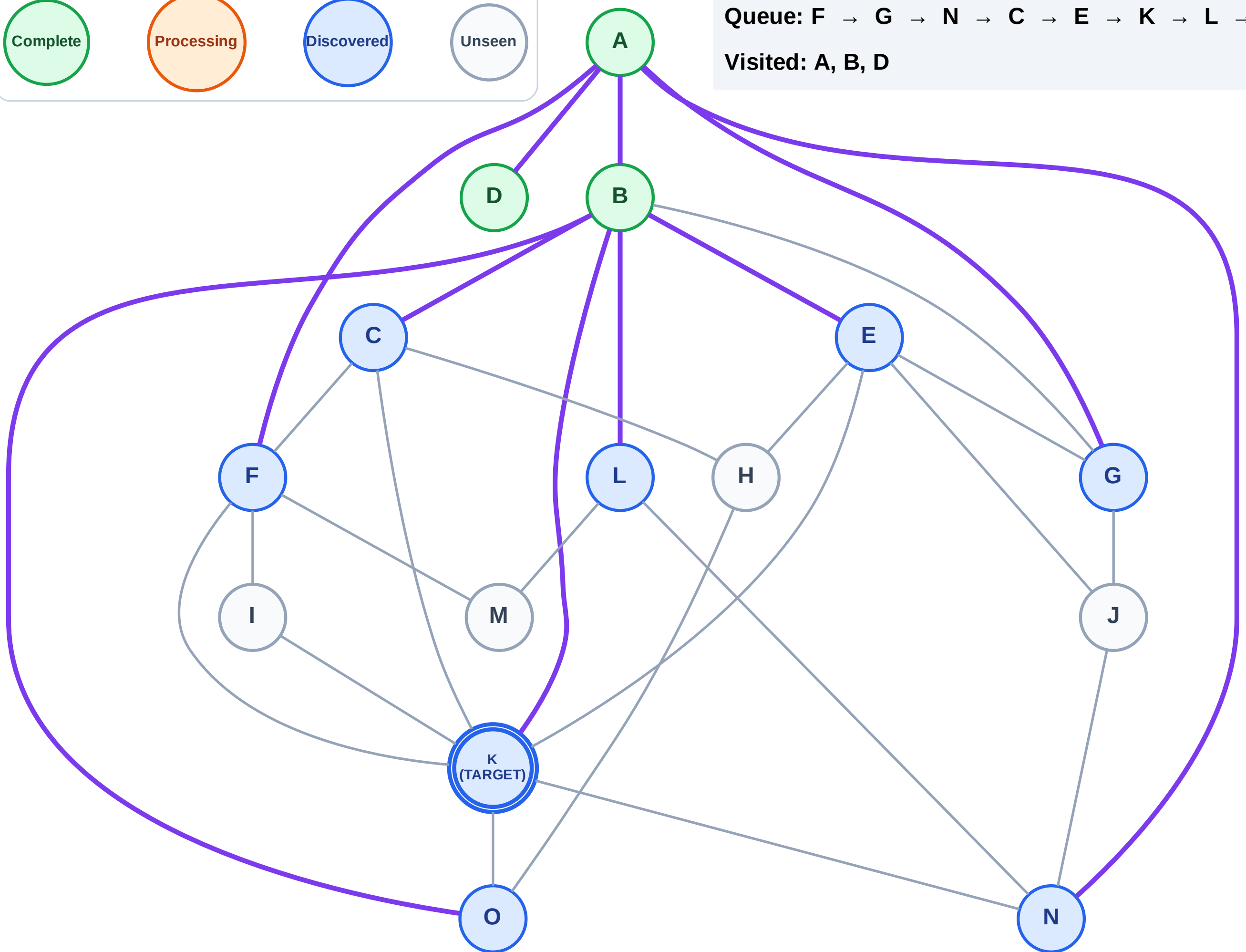
Step 7: No new neighbors from D

Legend



Queue: F → G → N → C → E → K → L → O

Visited: A, B, D



BFS Traversal

Step 8: Process node F

Legend

Complete

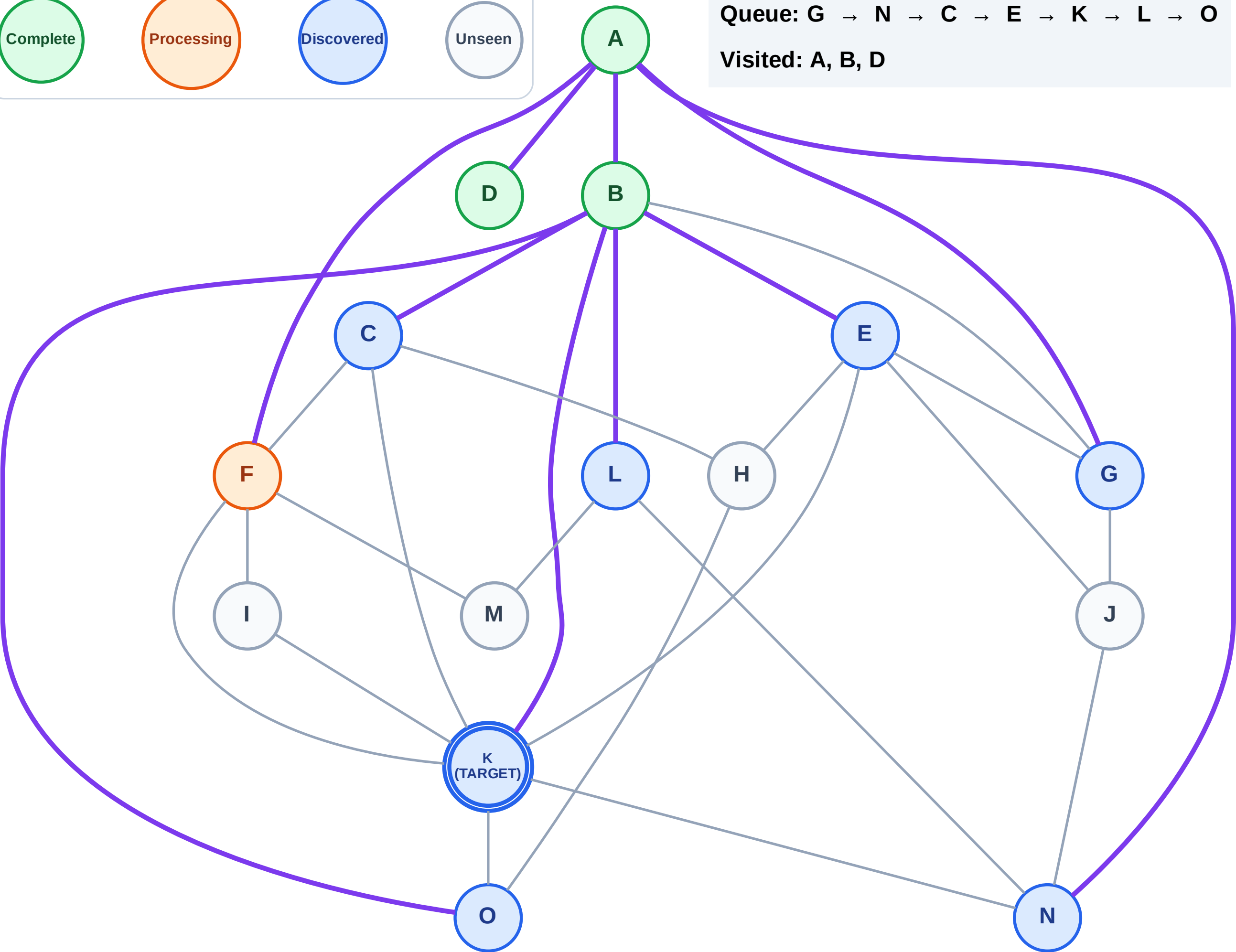
Processing

Discovered

Unseen

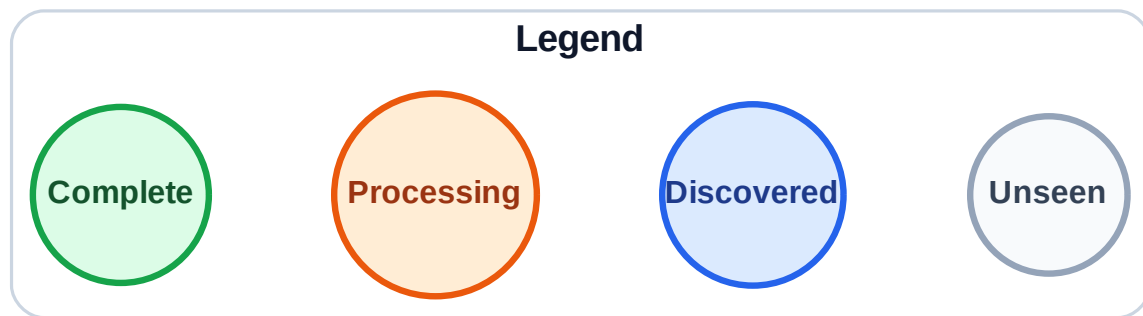
Queue: G → N → C → E → K → L → O

Visited: A, B, D



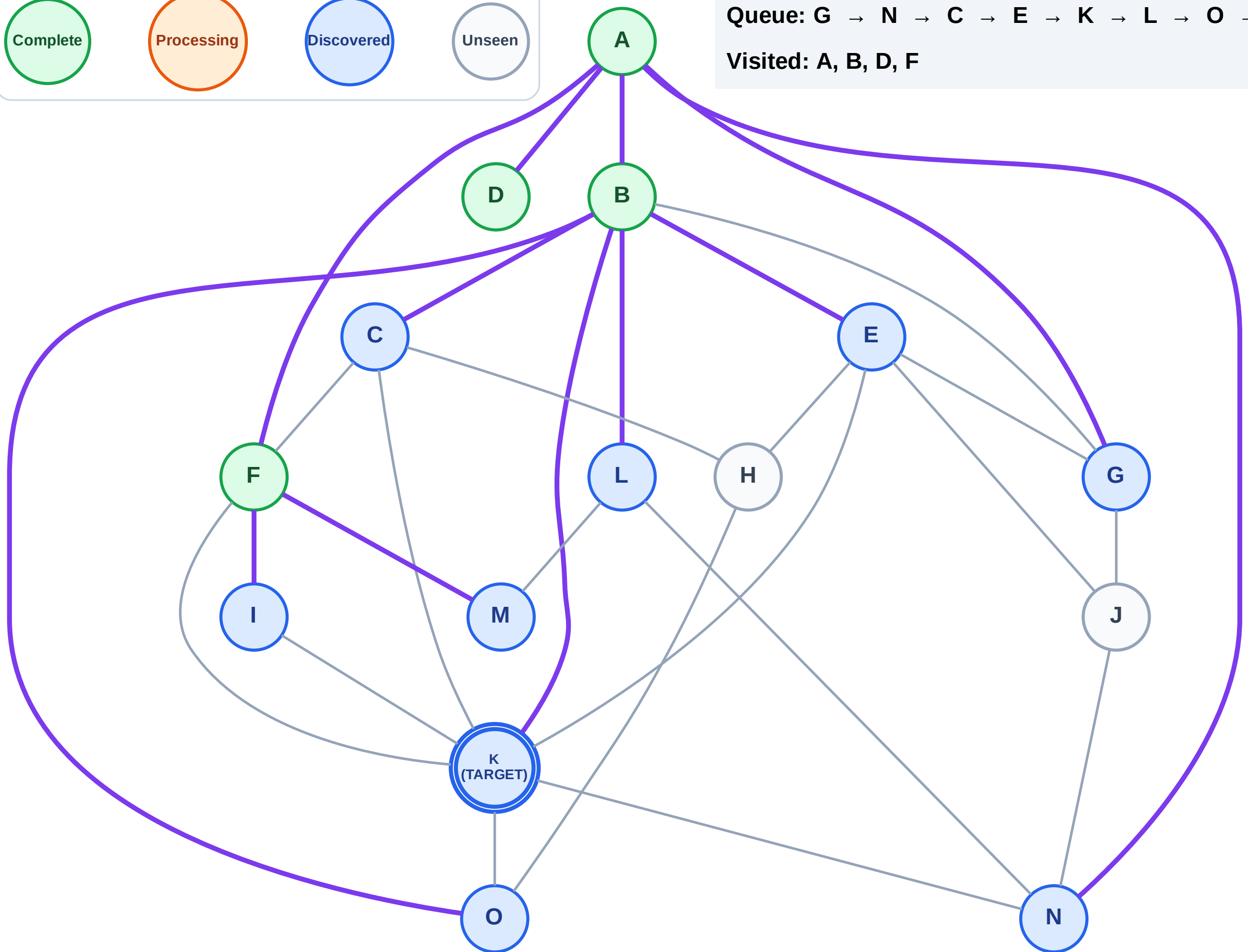
BFS Traversal

Step 9: Discover I, M from F



Queue: G → N → C → E → K → L → O → I → M

Visited: A, B, D, F



BFS Traversal

Step 10: Process node G

Legend

Complete

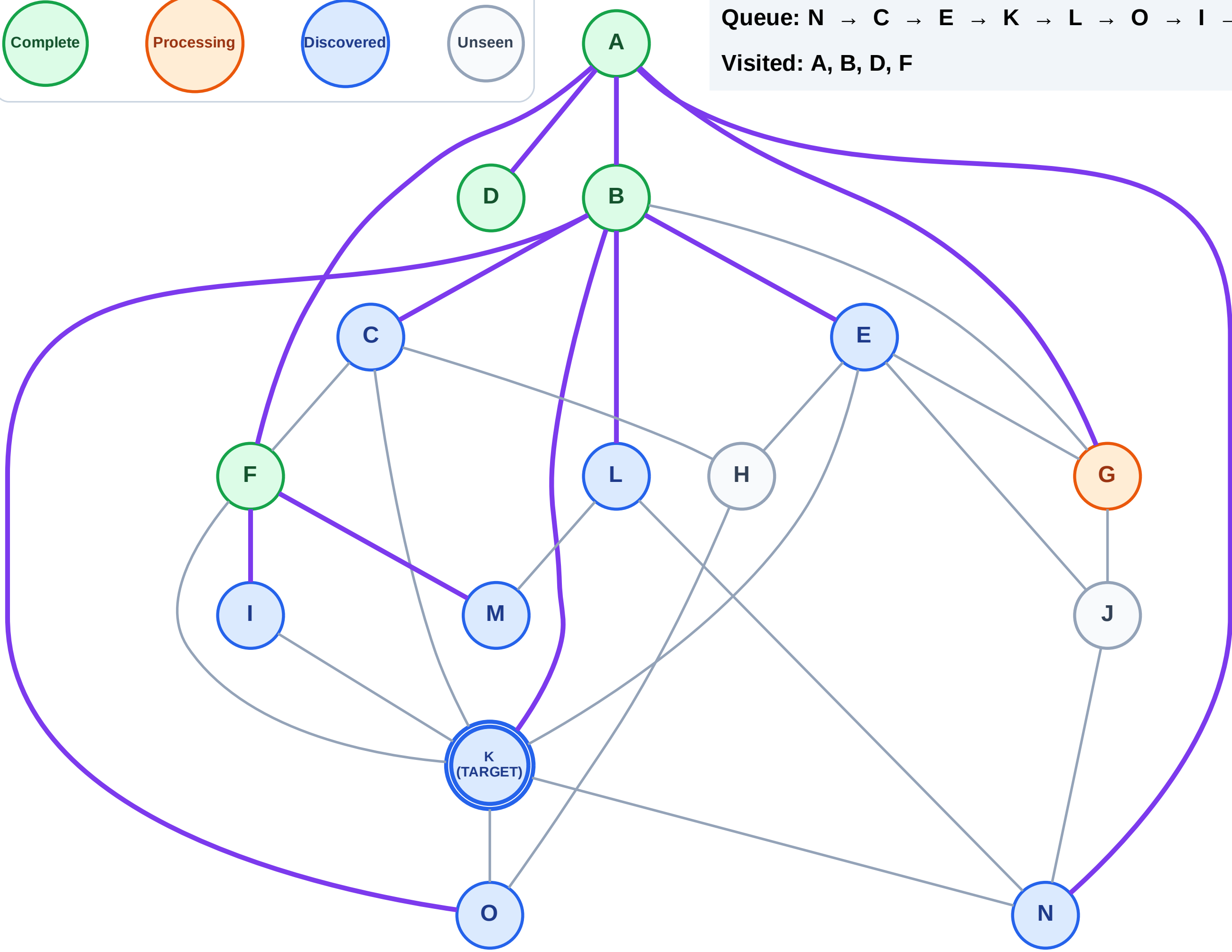
Processing

Discovered

Unseen

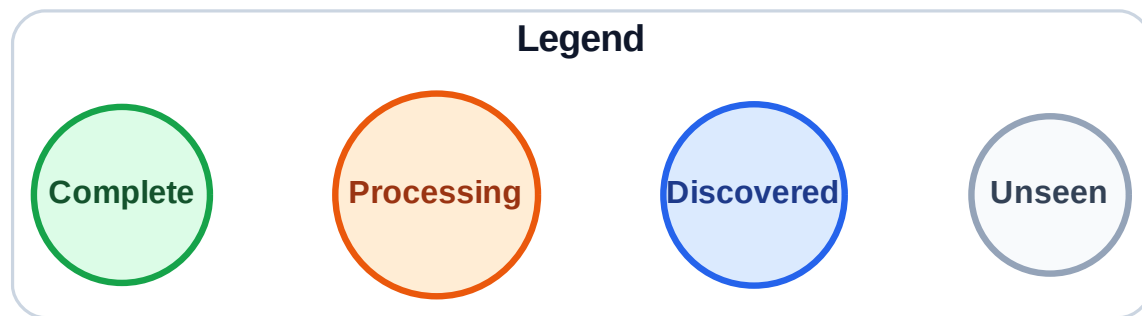
Queue: N → C → E → K → L → O → I → M

Visited: A, B, D, F



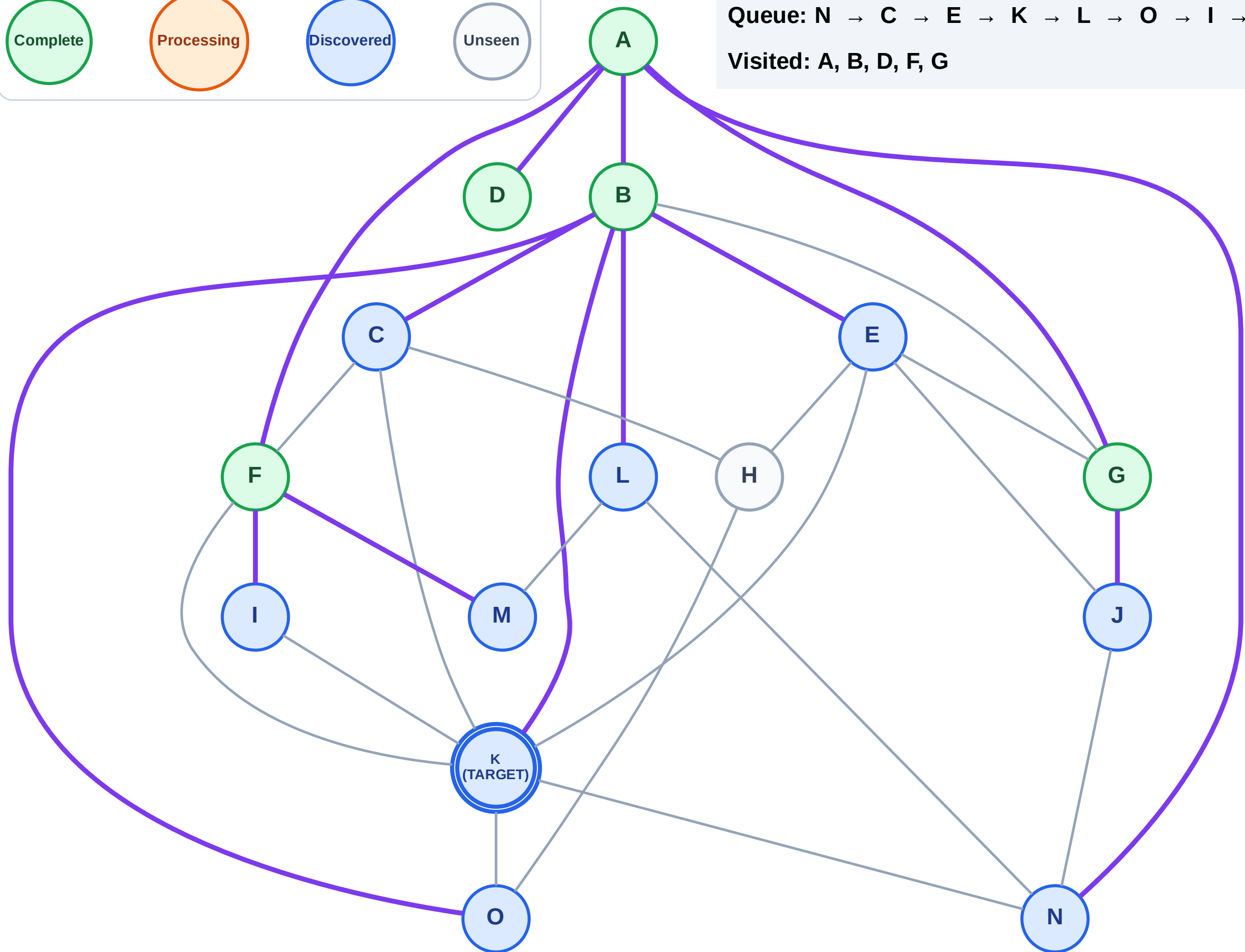
BFS Traversal

Step 11: Discover J from G



Queue: N → C → E → K → L → O → I → M → J

Visited: A, B, D, F, G



BFS Traversal

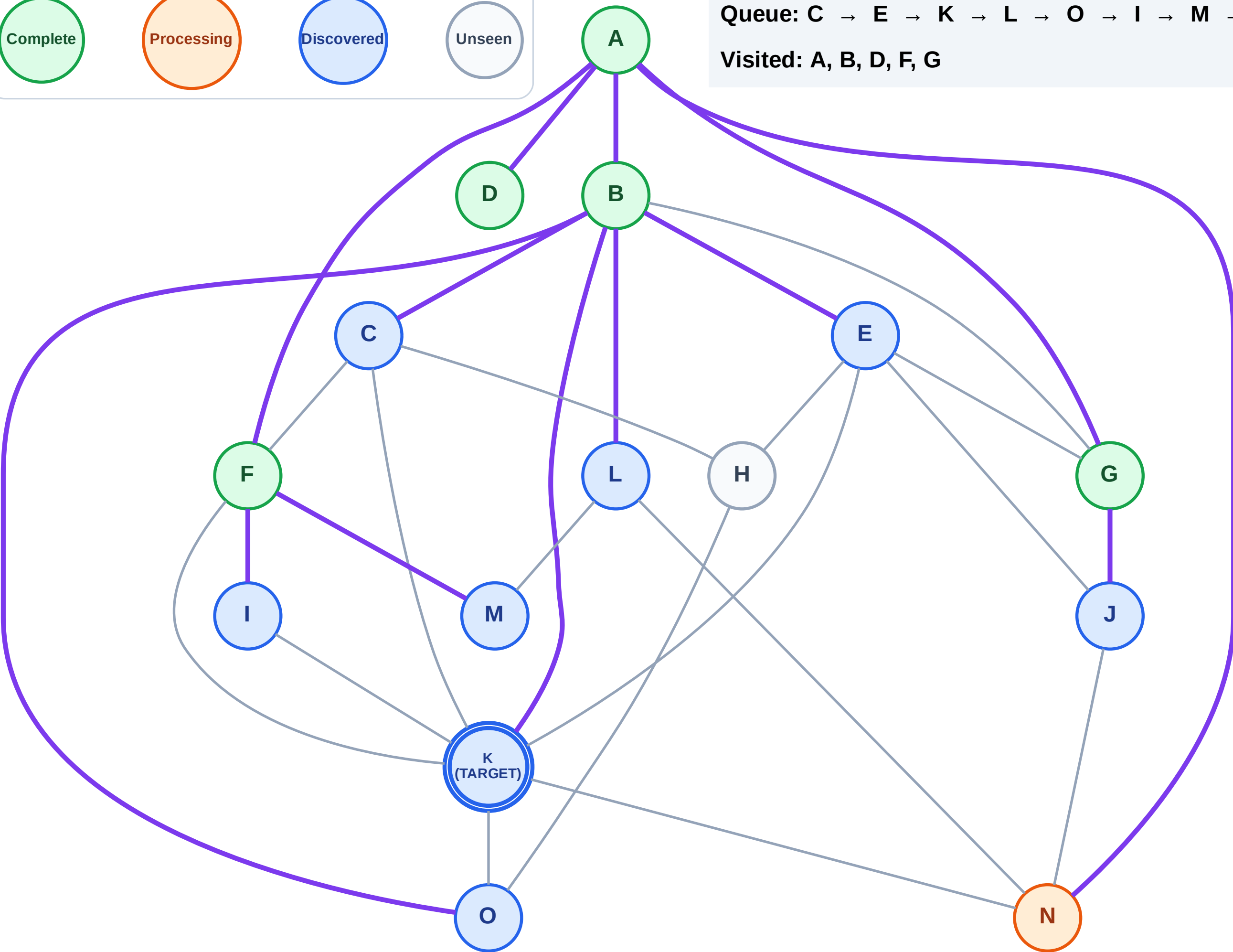
Step 12: Process node N

Legend



Queue: C → E → K → L → O → I → M → J

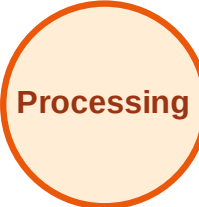
Visited: A, B, D, F, G



BFS Traversal

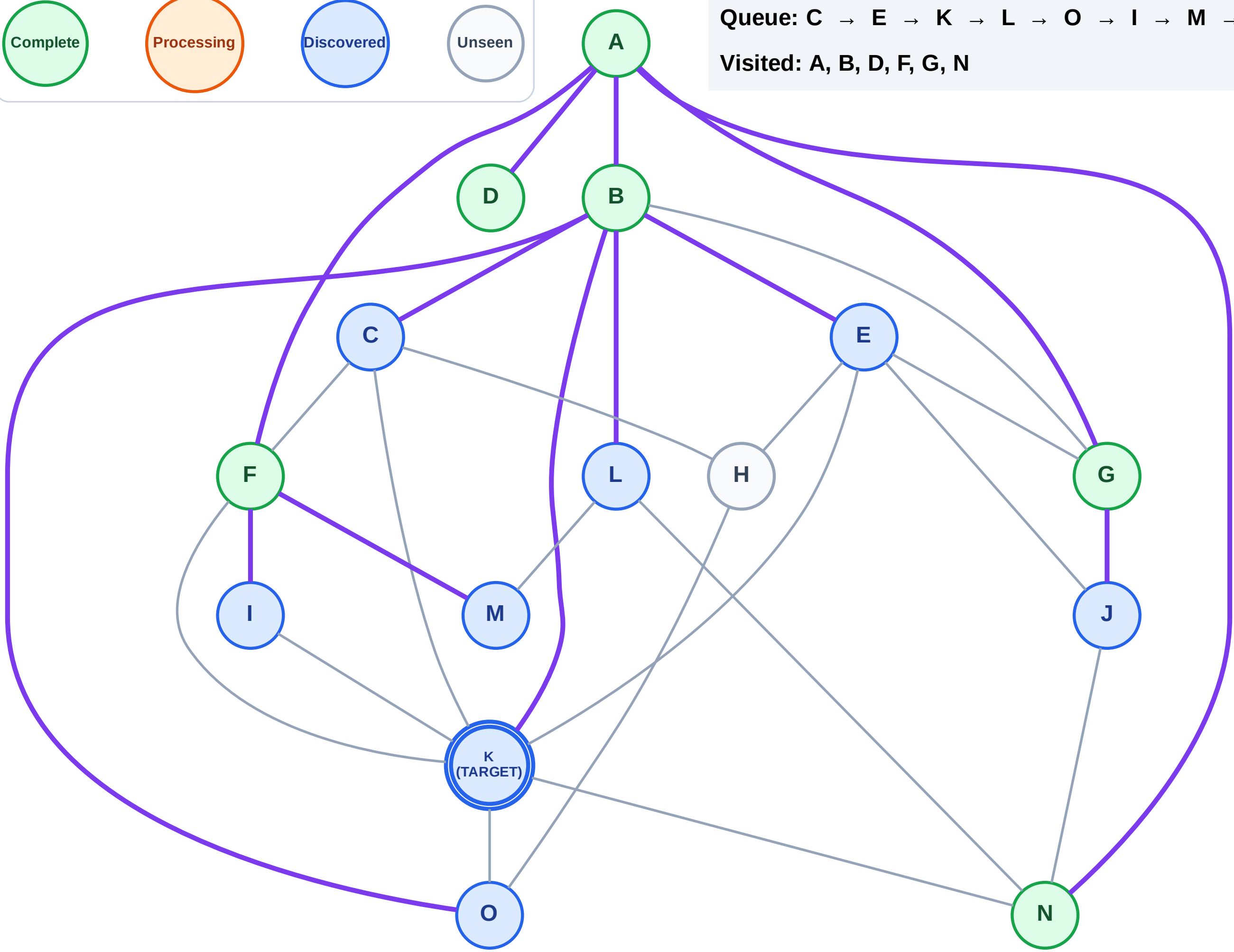
Step 13: No new neighbors from N

Legend



Queue: C → E → K → L → O → I → M → J

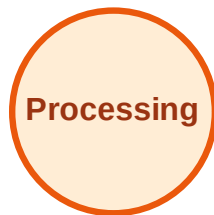
Visited: A, B, D, F, G, N



BFS Traversal

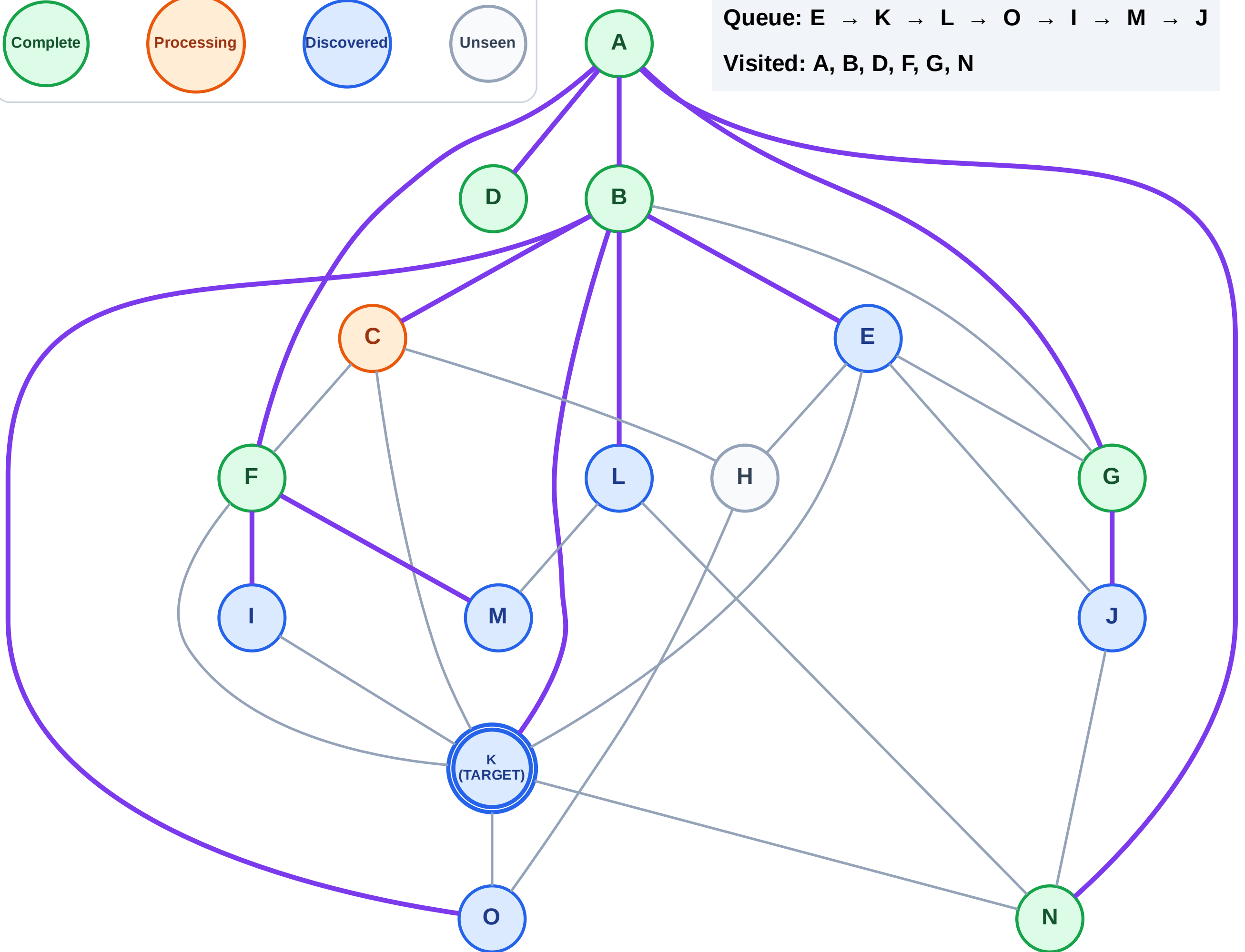
Step 14: Process node C

Legend



Queue: E → K → L → O → I → M → J

Visited: A, B, D, F, G, N



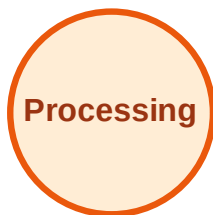
BFS Traversal

Step 15: Discover H from C

Legend



Complete



Processing



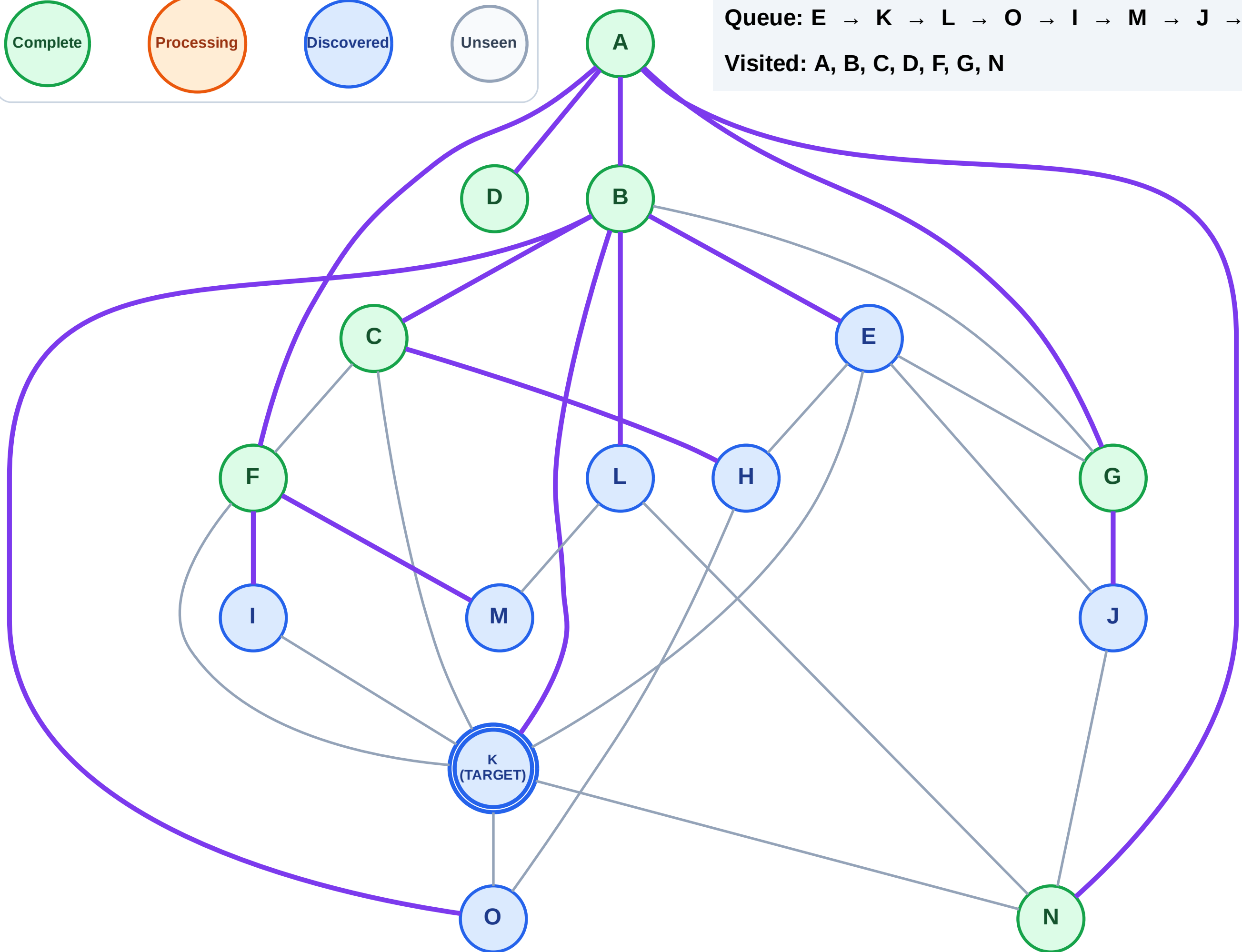
Discovered



Unseen

Queue: E → K → L → O → I → M → J → H

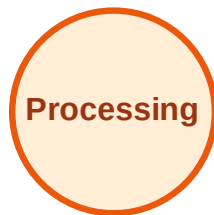
Visited: A, B, C, D, F, G, N



BFS Traversal

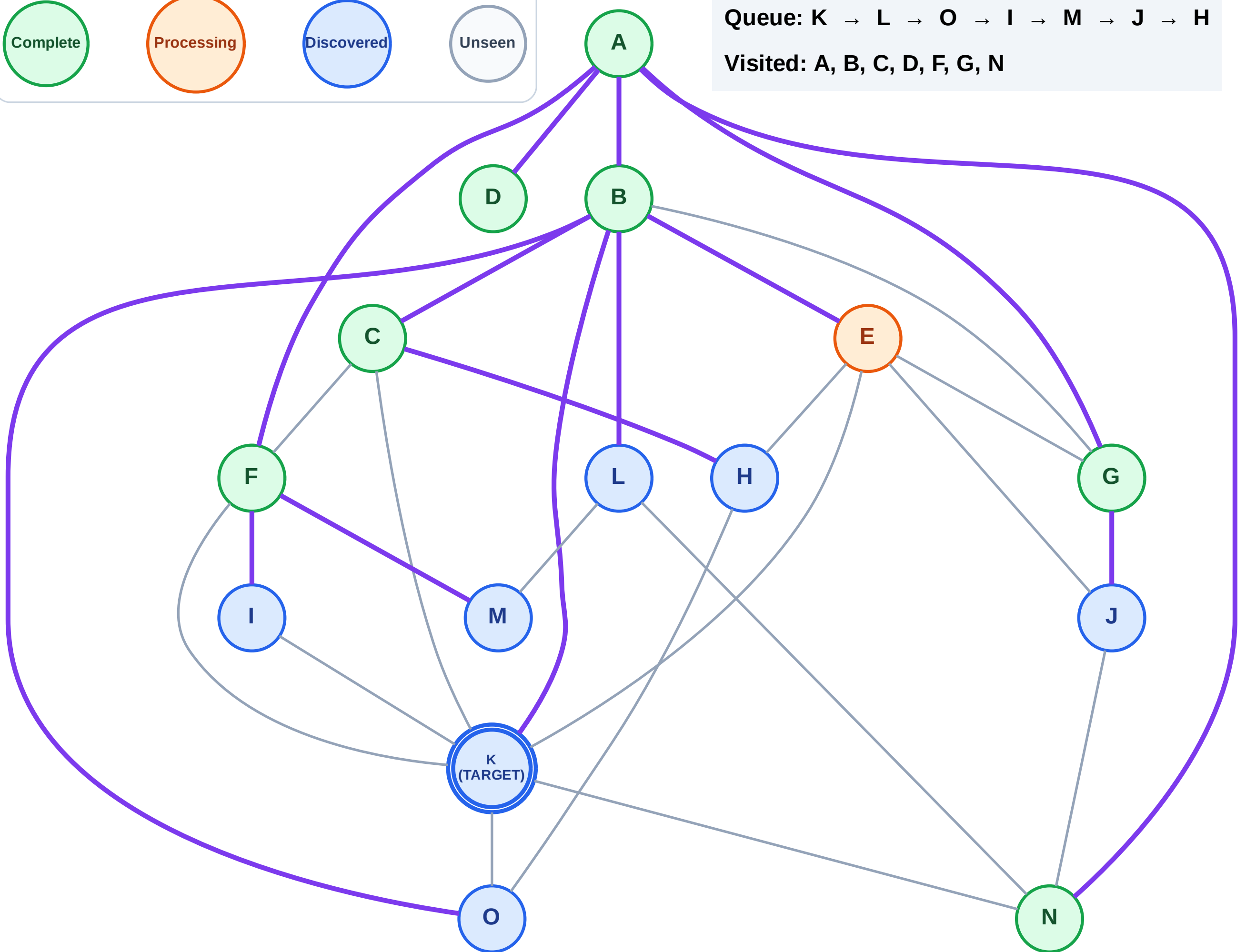
Step 16: Process node E

Legend



Queue: K → L → O → I → M → J → H

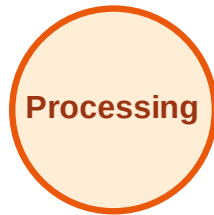
Visited: A, B, C, D, F, G, N



BFS Traversal

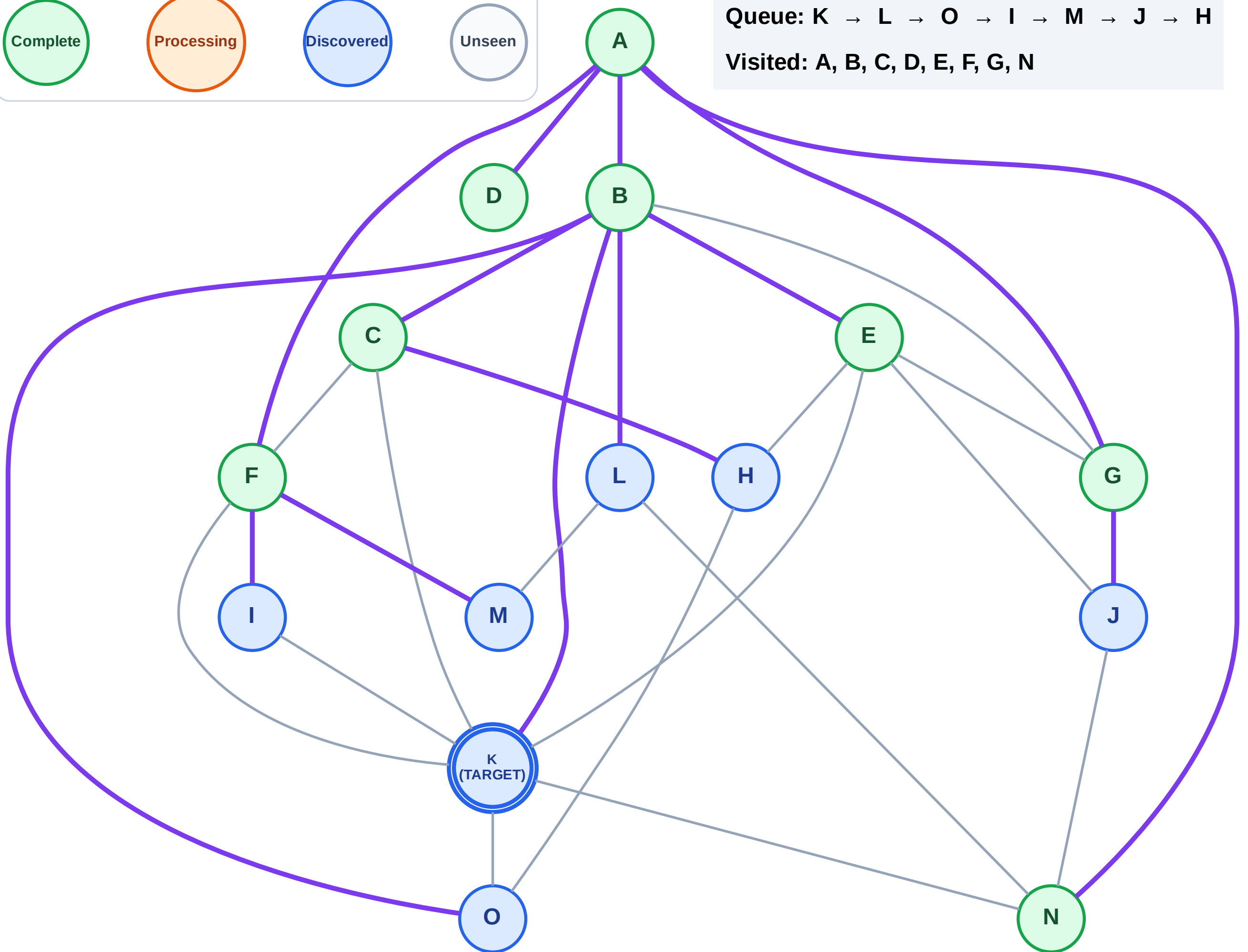
Step 17: No new neighbors from E

Legend



Queue: K → L → O → I → M → J → H

Visited: A, B, C, D, E, F, G, N



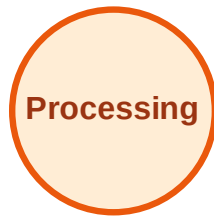
BFS Traversal

Step 18: Process node K

Legend



Complete



Processing



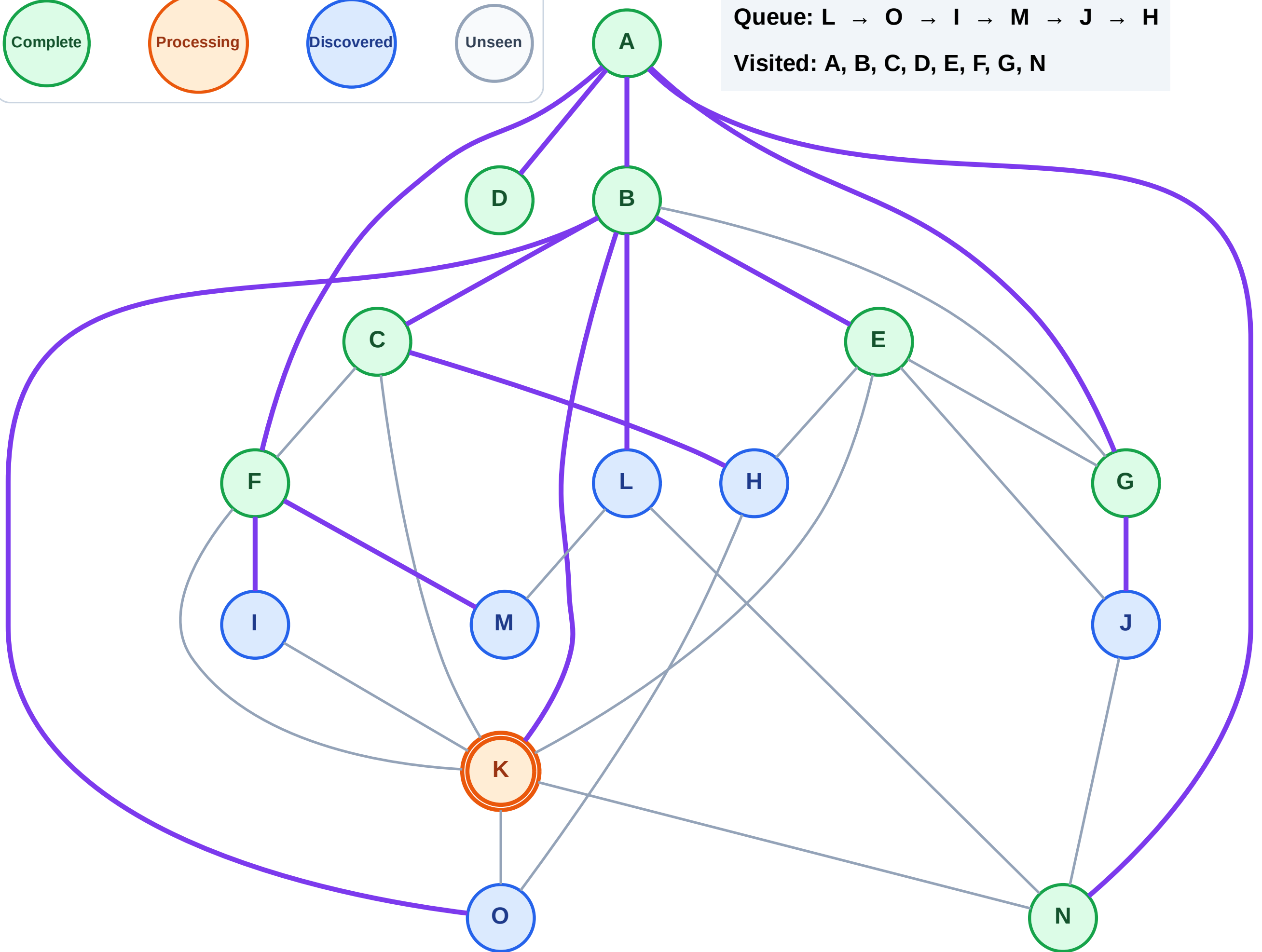
Discovered



Unseen

Queue: L → O → I → M → J → H

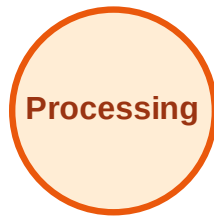
Visited: A, B, C, D, E, F, G, N



BFS Traversal

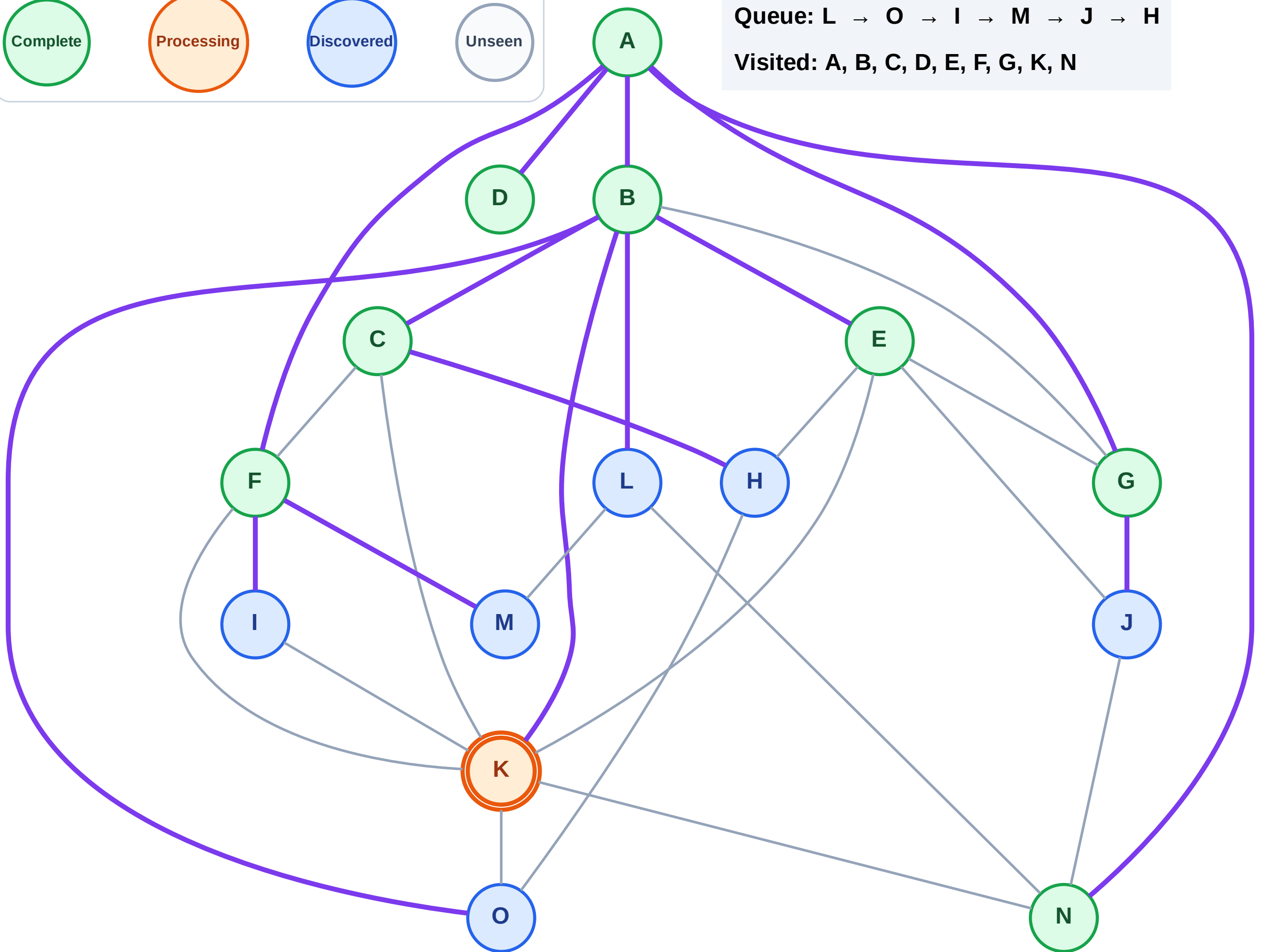
Step 19: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend



Queue: L → O → I → M → J → H

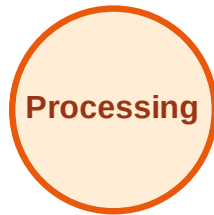
Visited: A, B, C, D, E, F, G, K, N



DFS Traversal

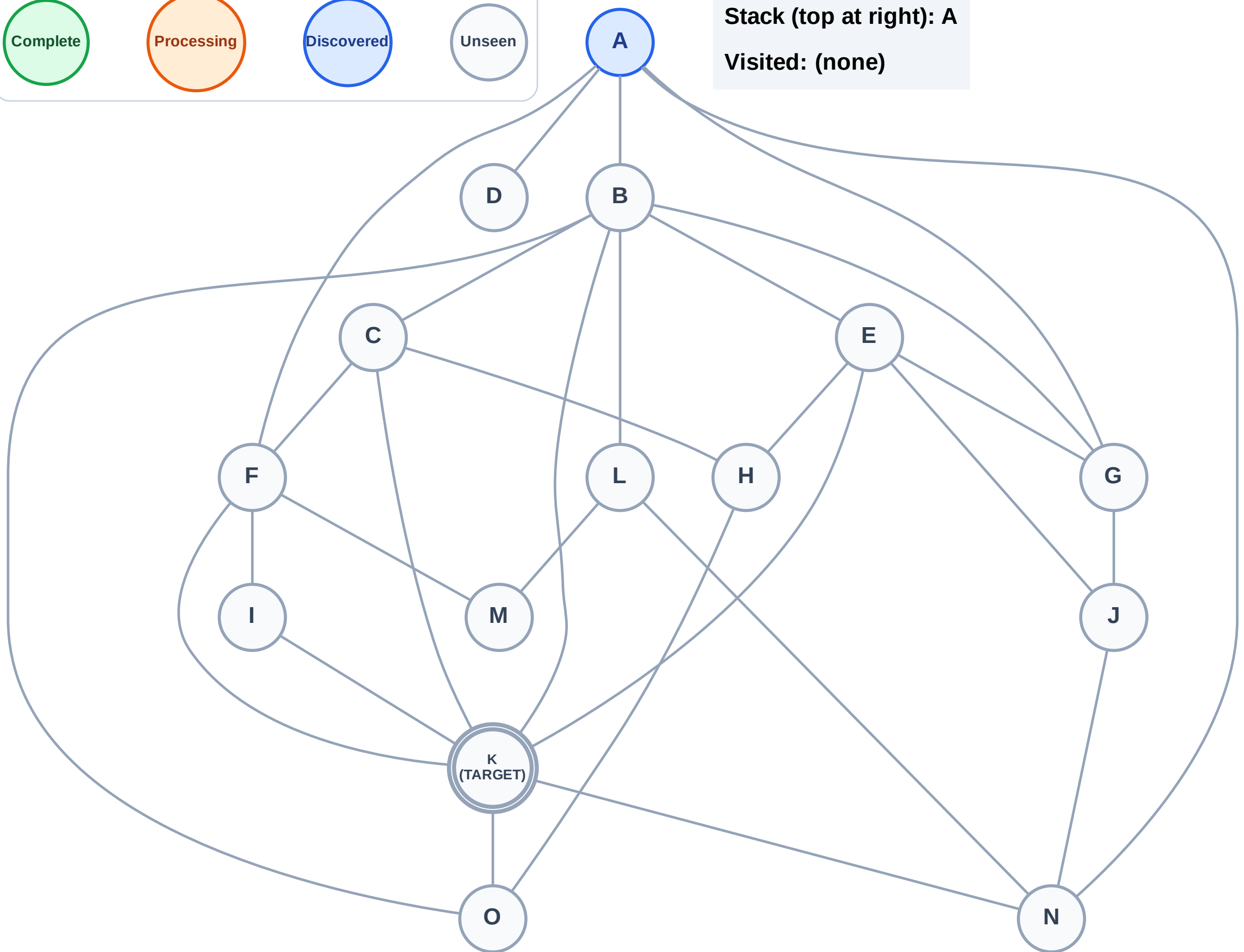
Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A

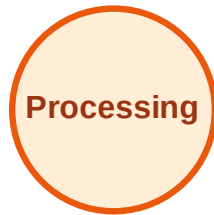
Visited: (none)



DFS Traversal

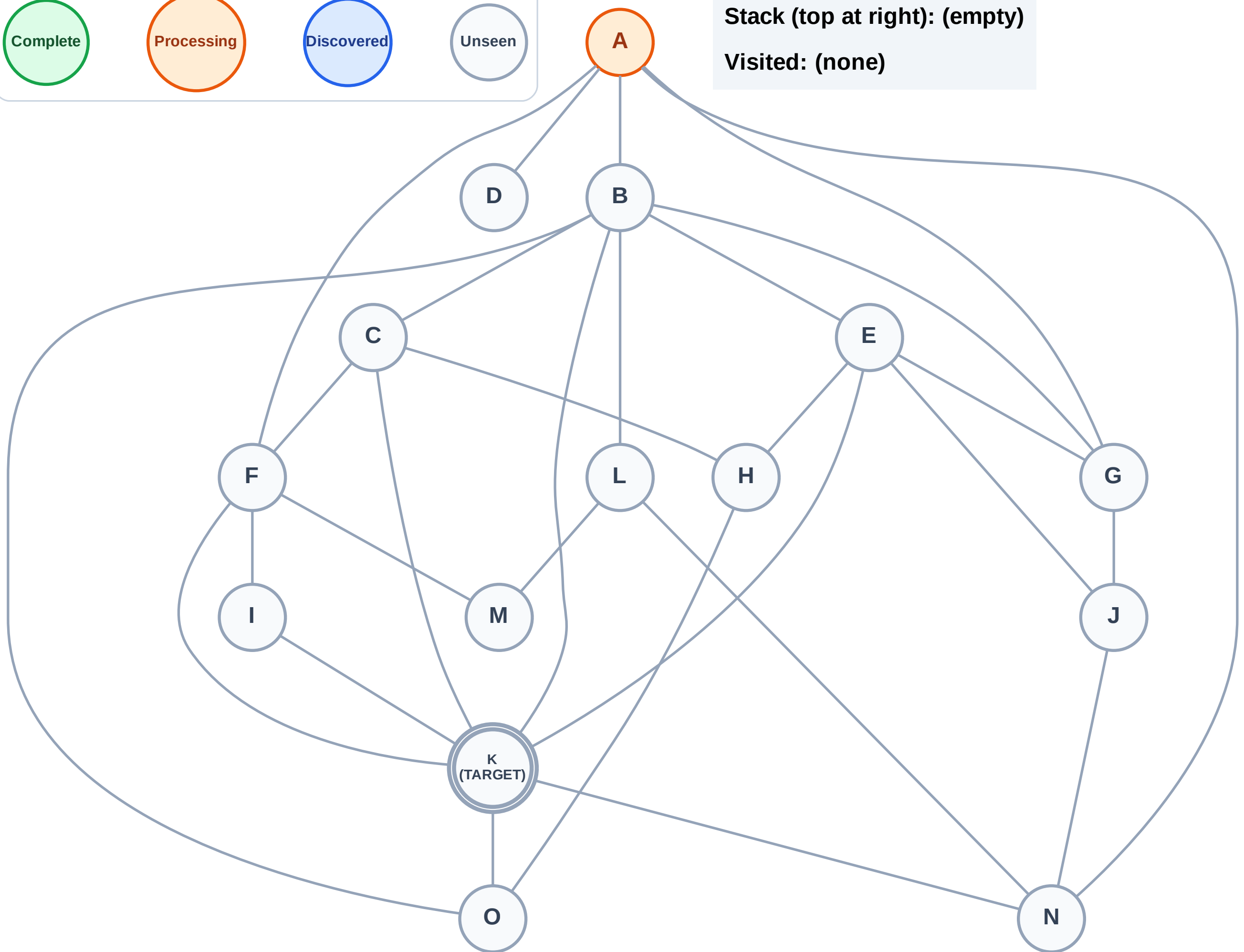
Step 2: Process node A

Legend



Stack (top at right): (empty)

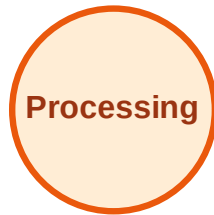
Visited: (none)



DFS Traversal

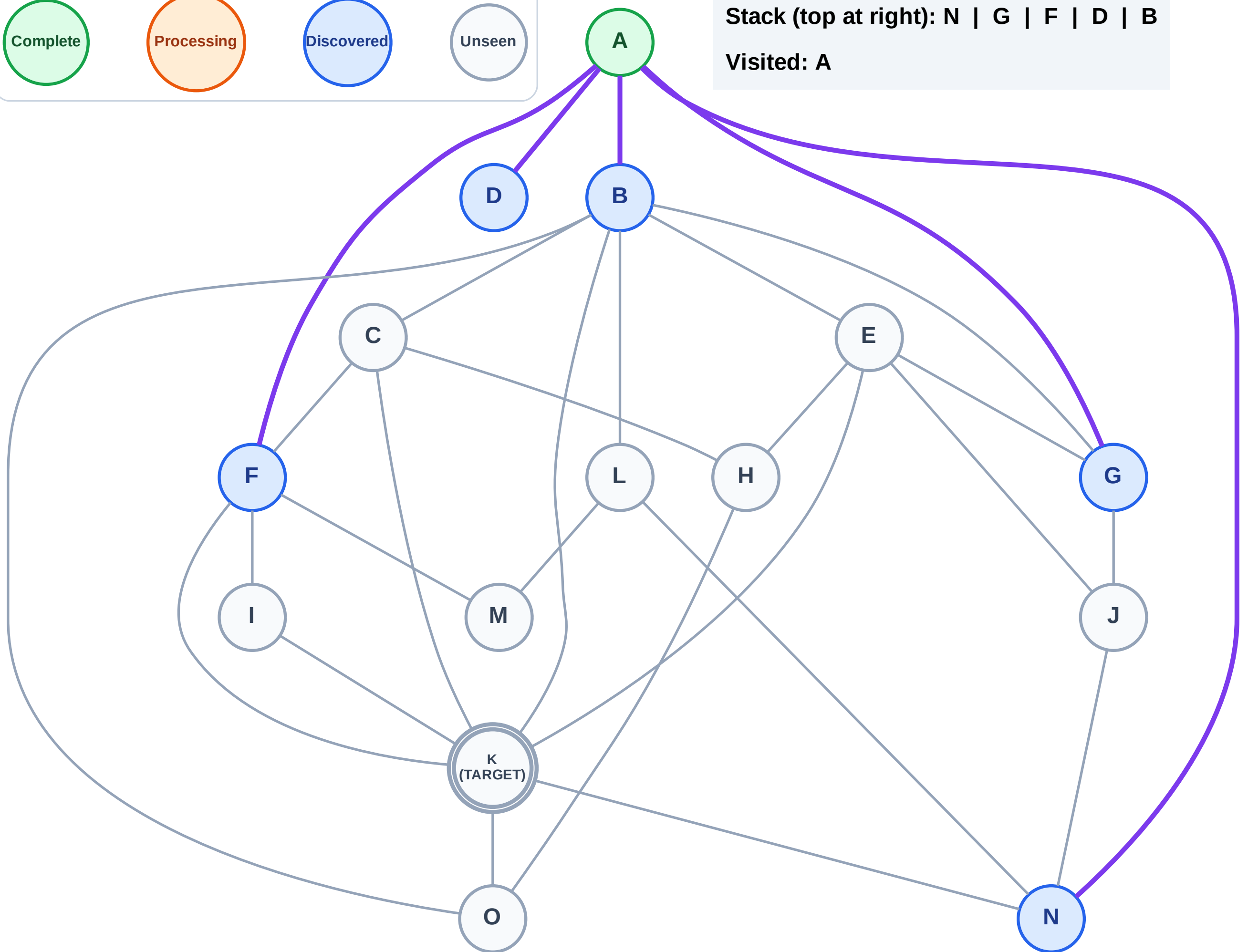
Step 3: Discover N, G, F, D, B from A

Legend



Stack (top at right): N | G | F | D | B

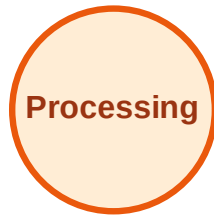
Visited: A



DFS Traversal

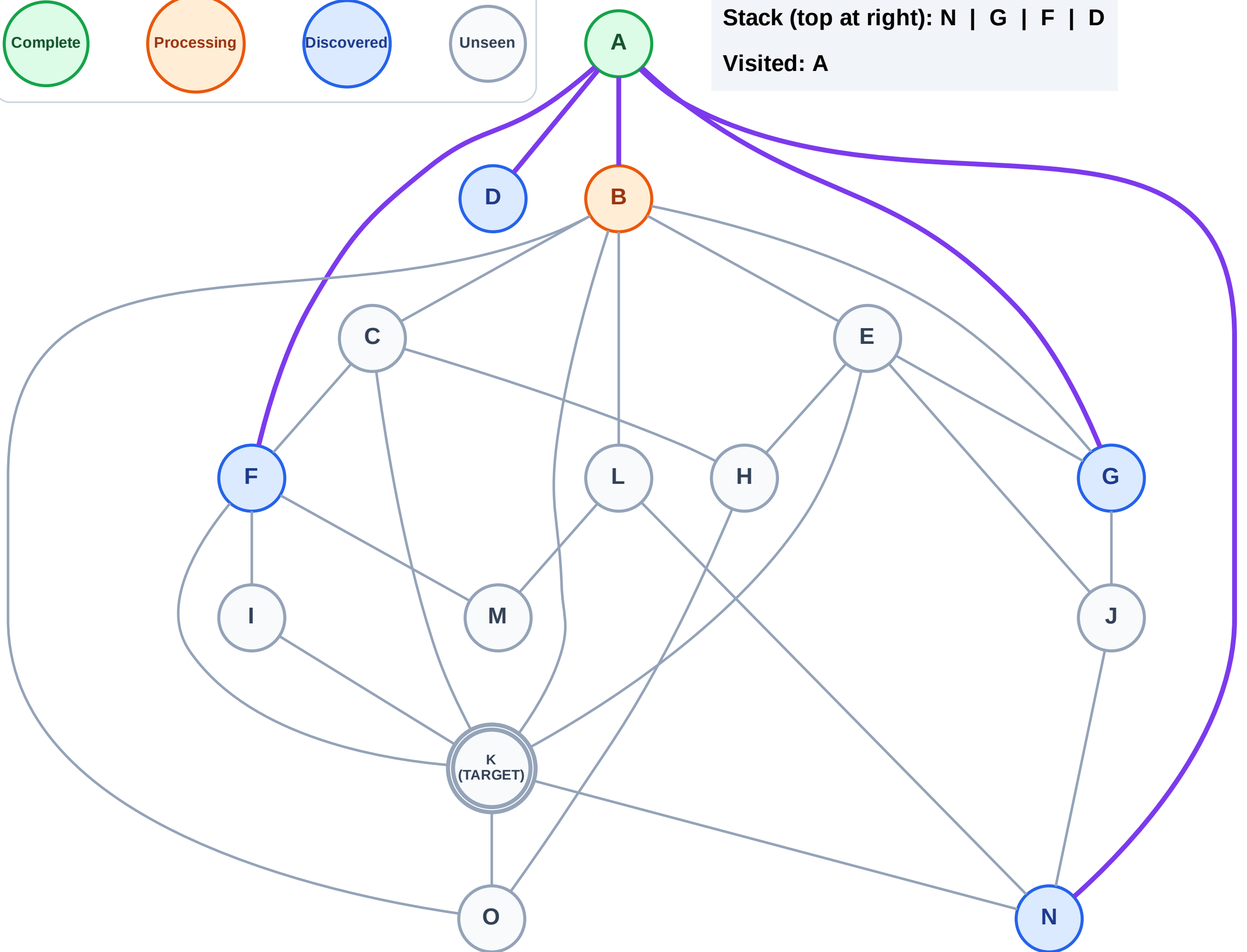
Step 4: Process node B

Legend



Stack (top at right): N | G | F | D

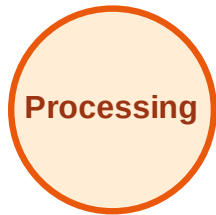
Visited: A



DFS Traversal

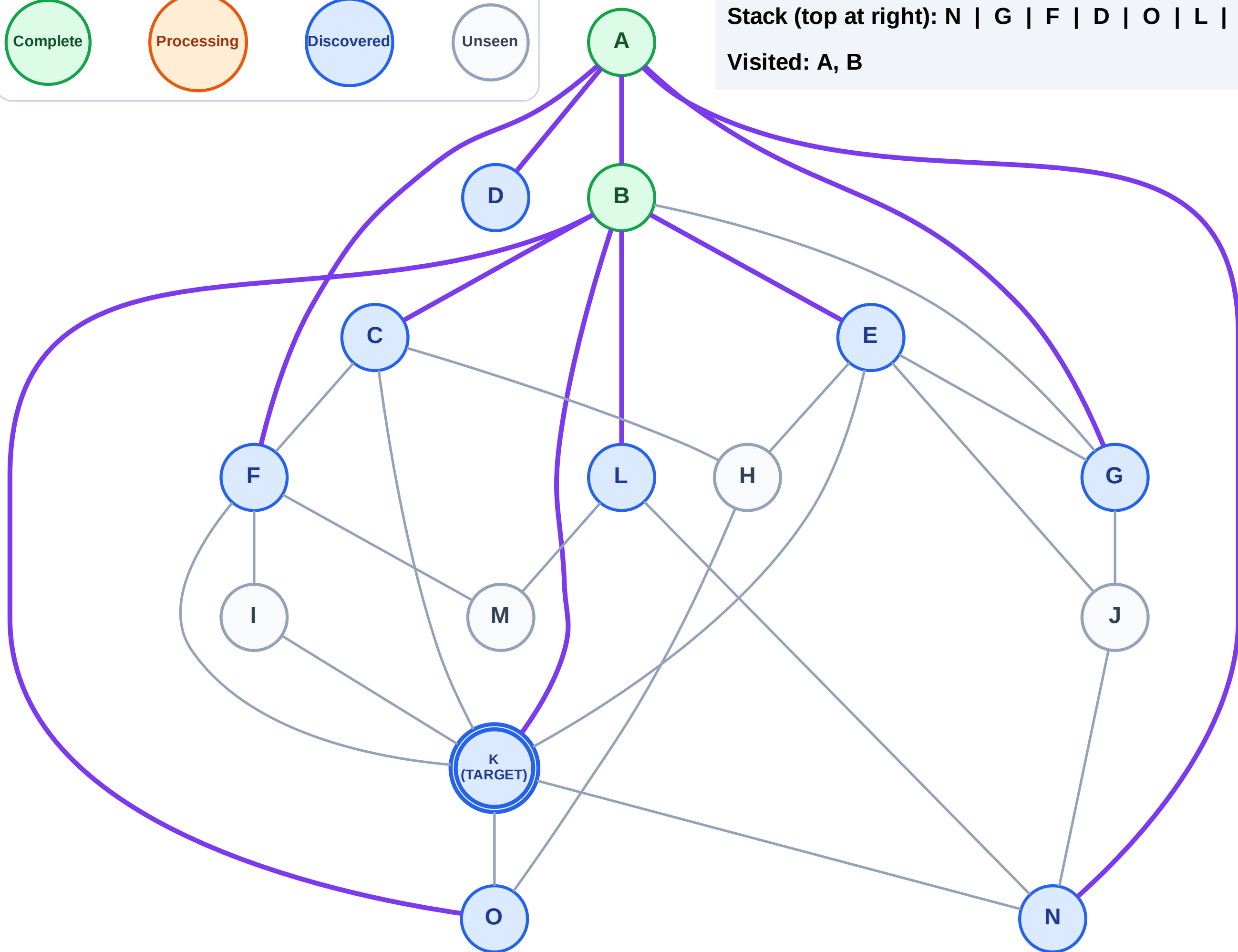
Step 5: Discover O, L, K, E, C from B

Legend



Stack (top at right): N | G | F | D | O | L | K | E | C

Visited: A, B



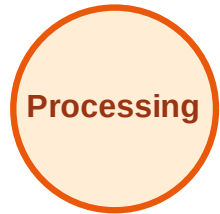
DFS Traversal

Step 6: Process node C

Legend



Complete



Processing



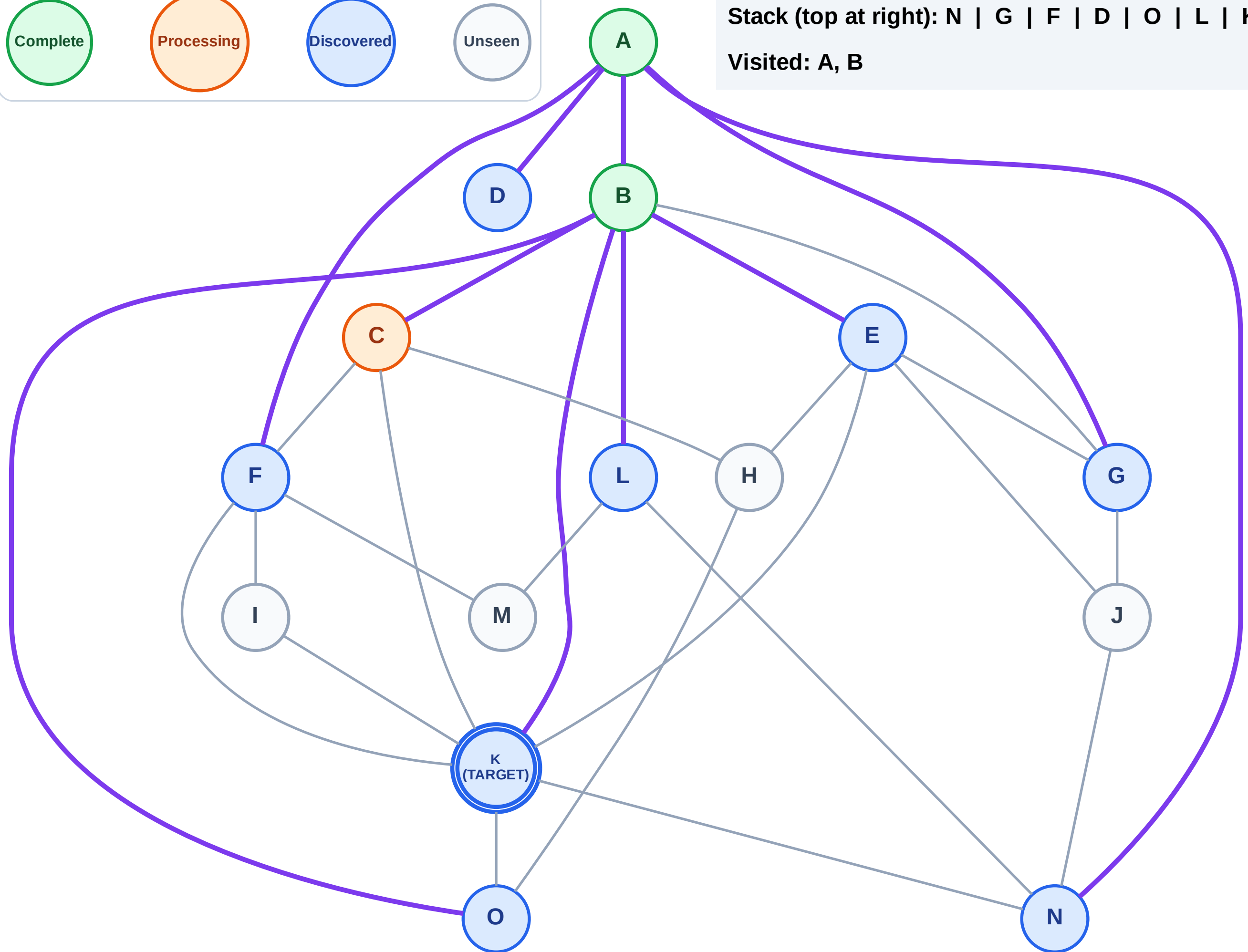
Discovered



Unseen

Stack (top at right): N | G | F | D | O | L | K | E

Visited: A, B



DFS Traversal

Step 7: Discover H from C

Legend

Complete

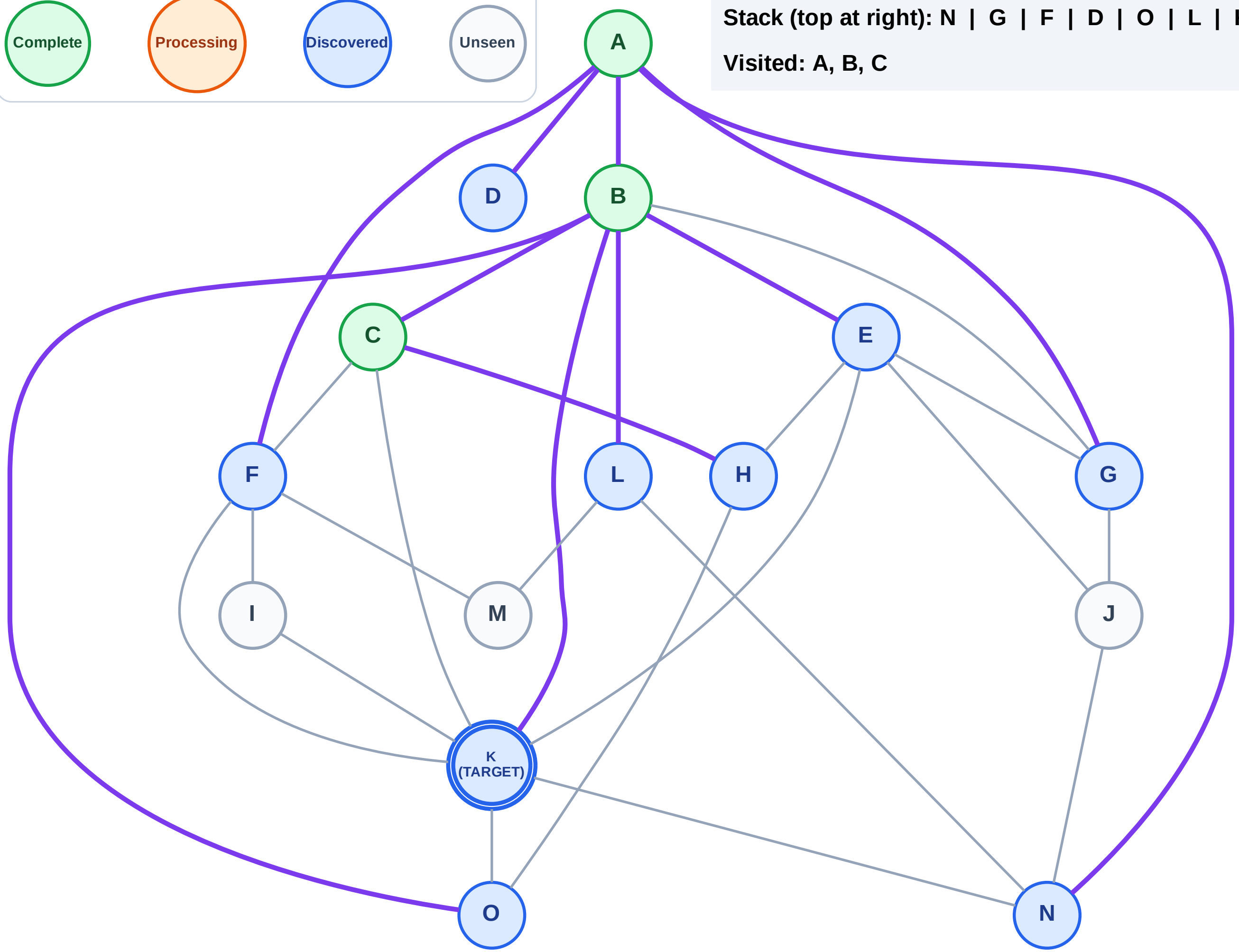
Processing

Discovered

Unseen

Stack (top at right): N | G | F | D | O | L | K | E | H

Visited: A, B, C



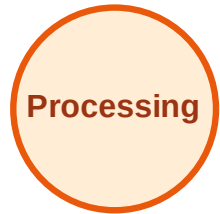
DFS Traversal

Step 8: Process node H

Legend



Complete



Processing



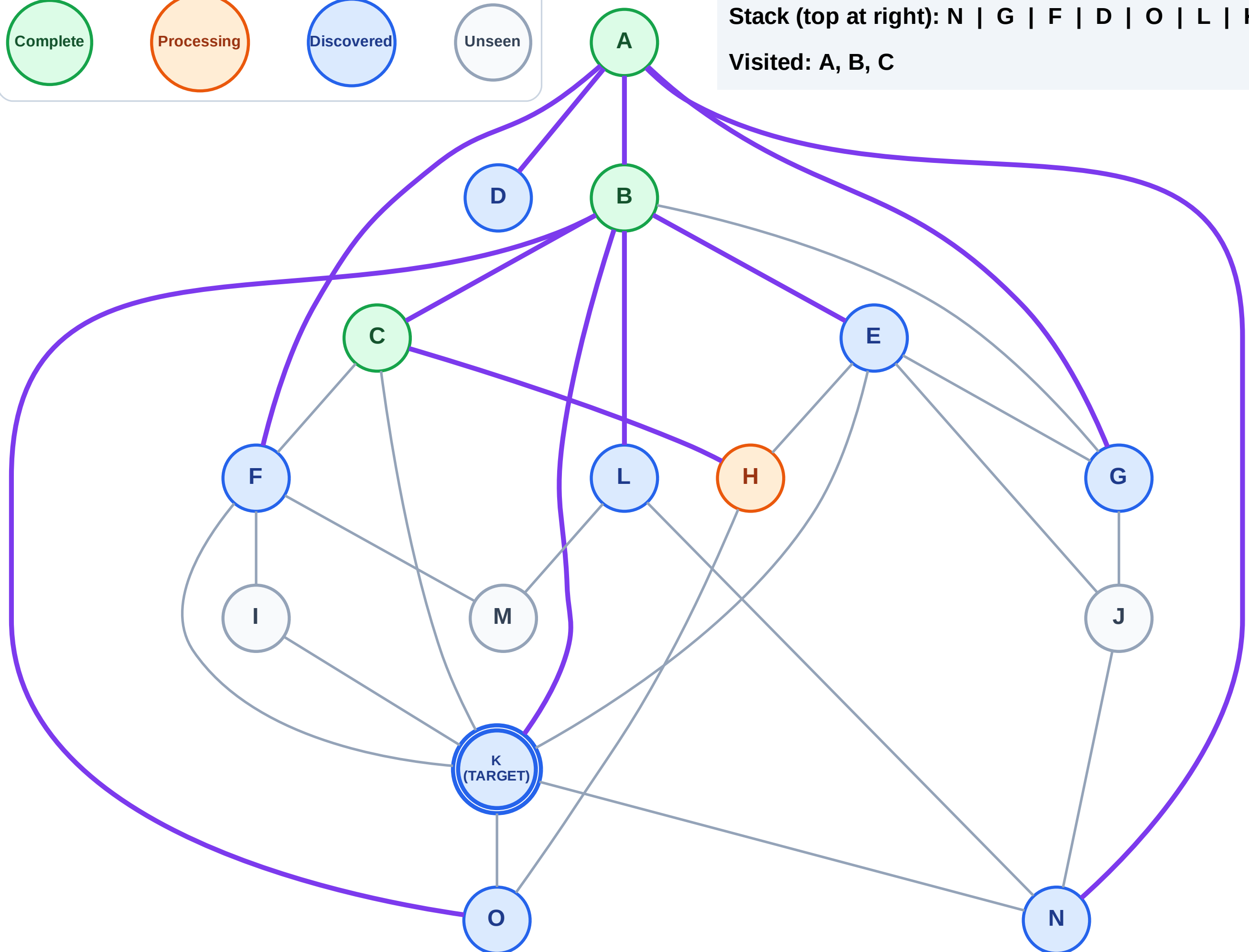
Discovered



Unseen

Stack (top at right): N | G | F | D | O | L | K | E

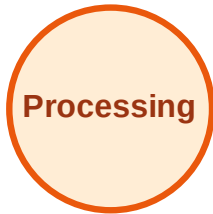
Visited: A, B, C



DFS Traversal

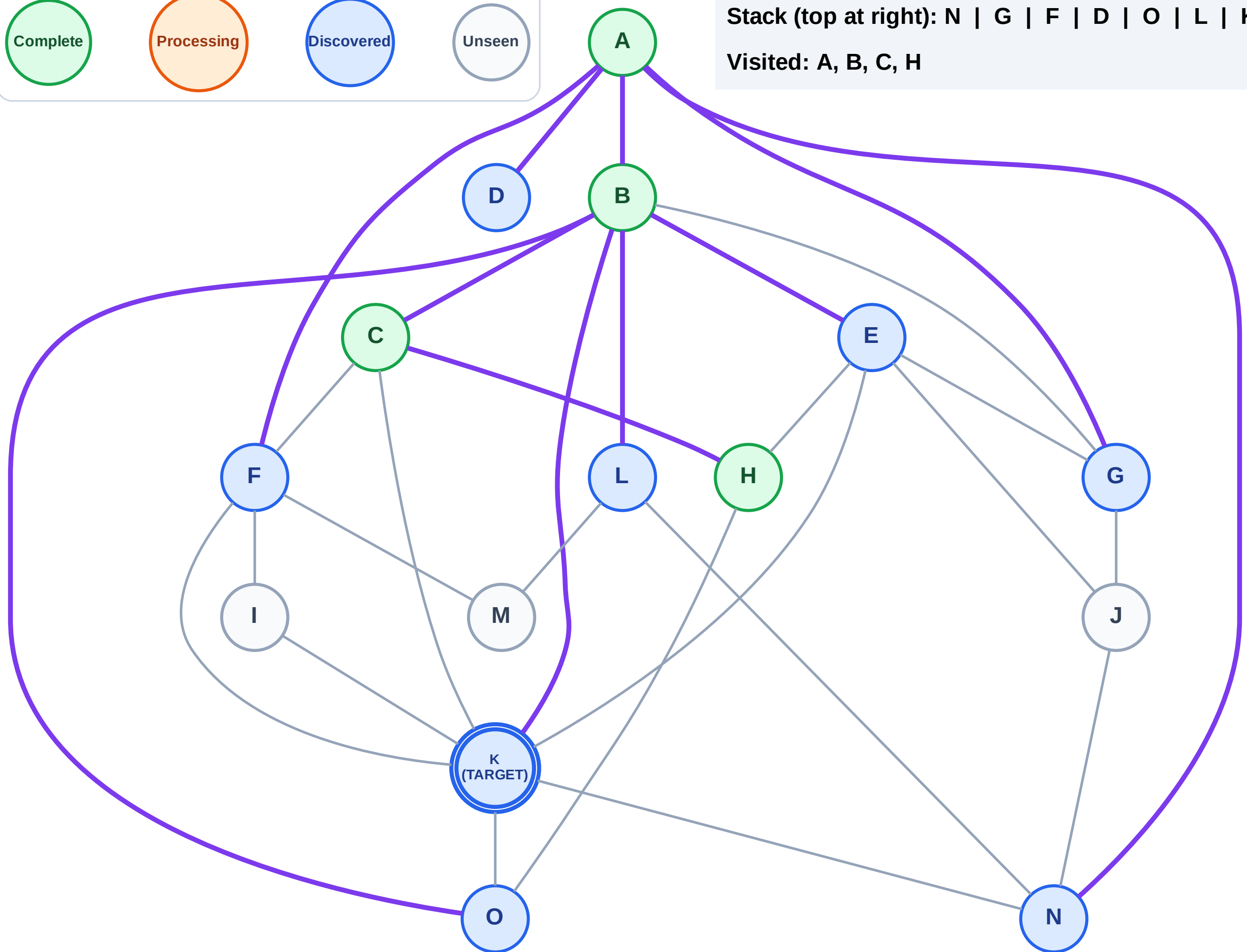
Step 9: No new neighbors from H

Legend



Stack (top at right): N | G | F | D | O | L | K | E

Visited: A, B, C, H



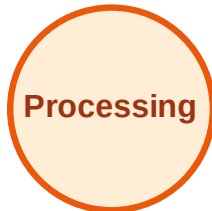
DFS Traversal

Step 10: Process node E

Legend



Complete



Processing



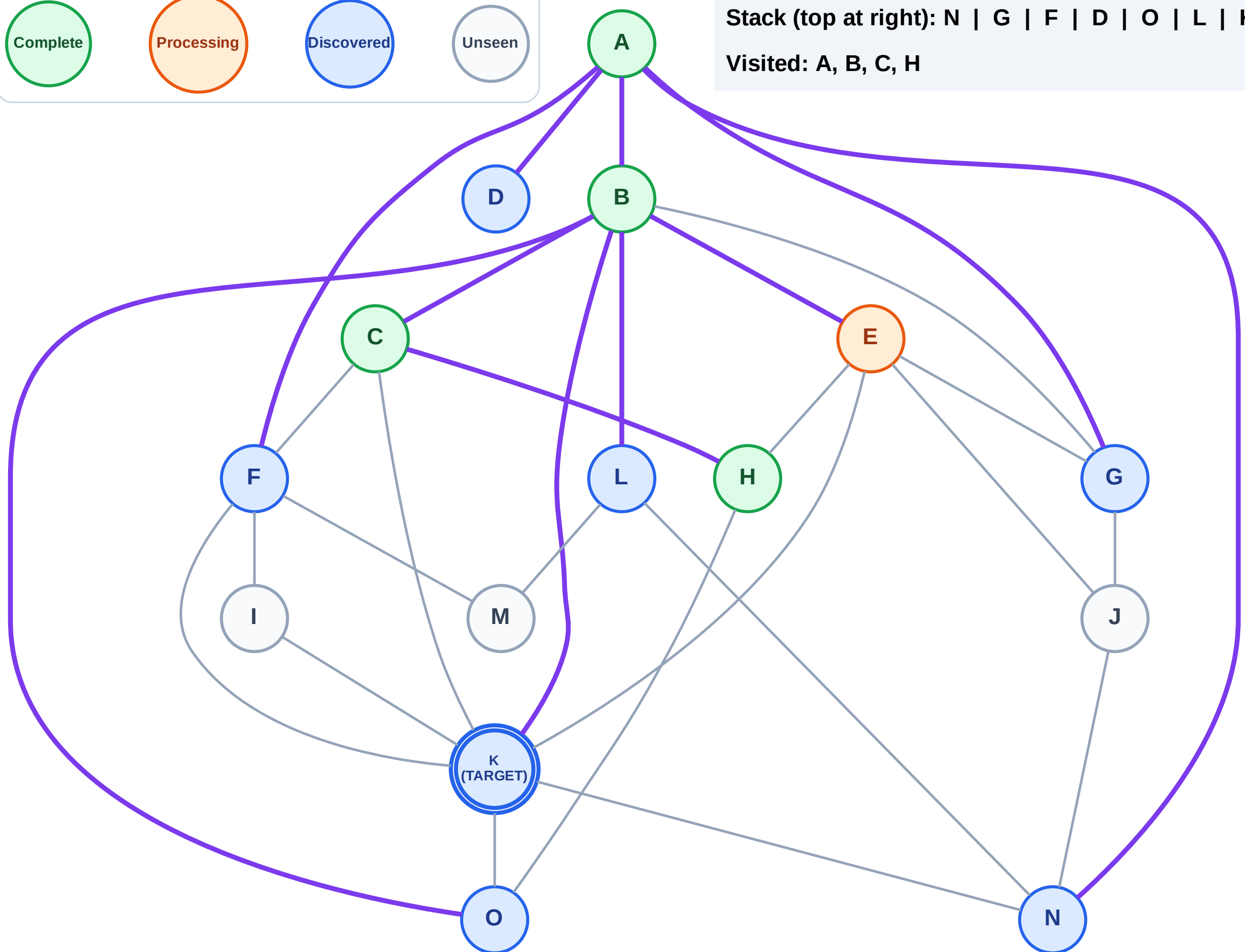
Discovered



Unseen

Stack (top at right): N | G | F | D | O | L | K

Visited: A, B, C, H



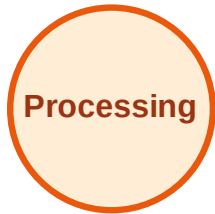
DFS Traversal

Step 11: Discover J from E

Legend



Complete



Processing



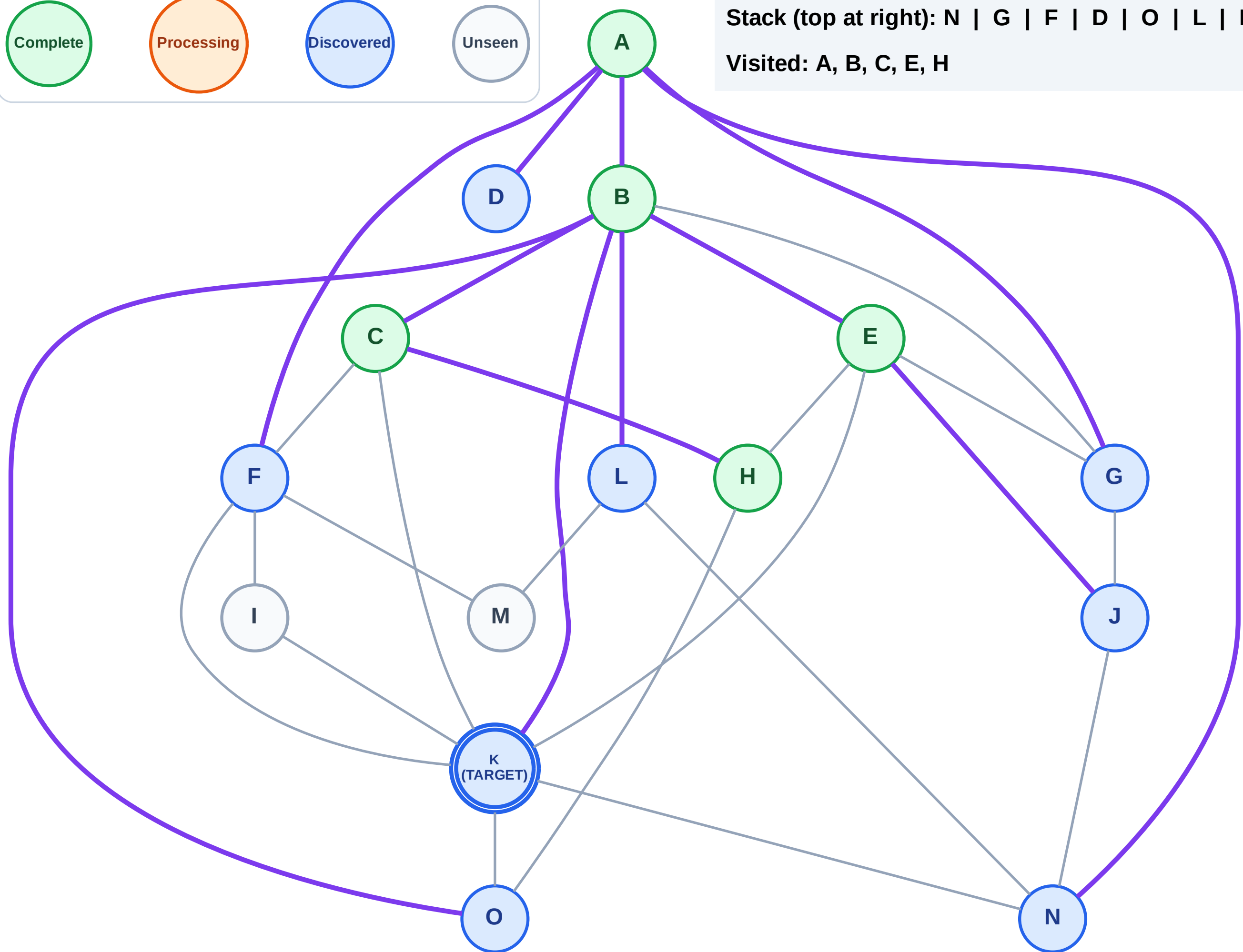
Discovered



Unseen

Stack (top at right): N | G | F | D | O | L | K | J

Visited: A, B, C, E, H



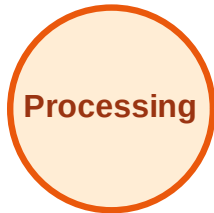
DFS Traversal

Step 12: Process node J

Legend



Complete



Processing



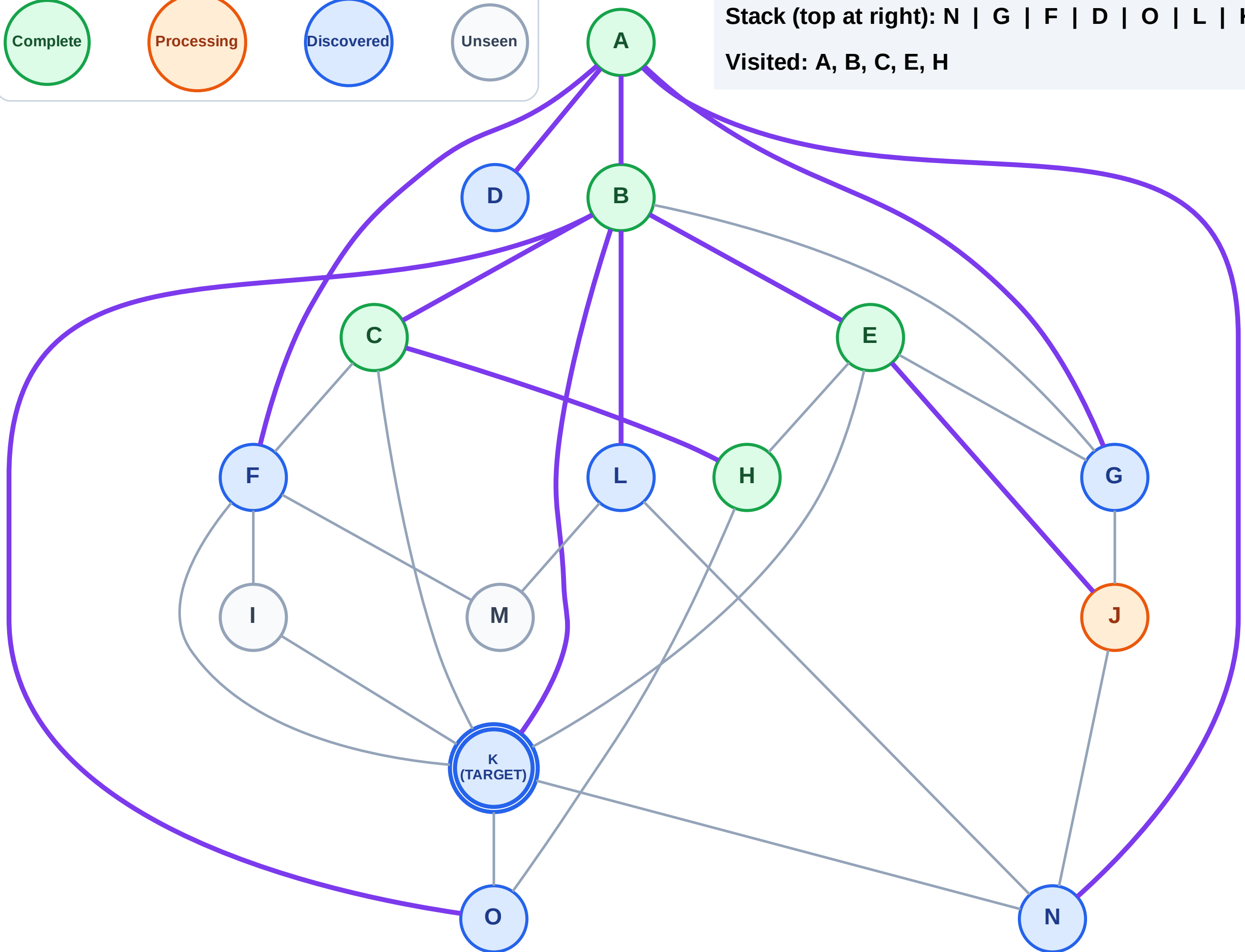
Discovered



Unseen

Stack (top at right): N | G | F | D | O | L | K

Visited: A, B, C, E, H



Step 13: No new neighbors from J

Stack (top at right): N | G | F | D | O | L | K

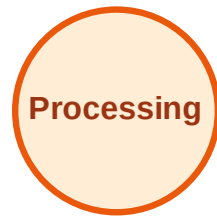
Unseen



DFS Traversal

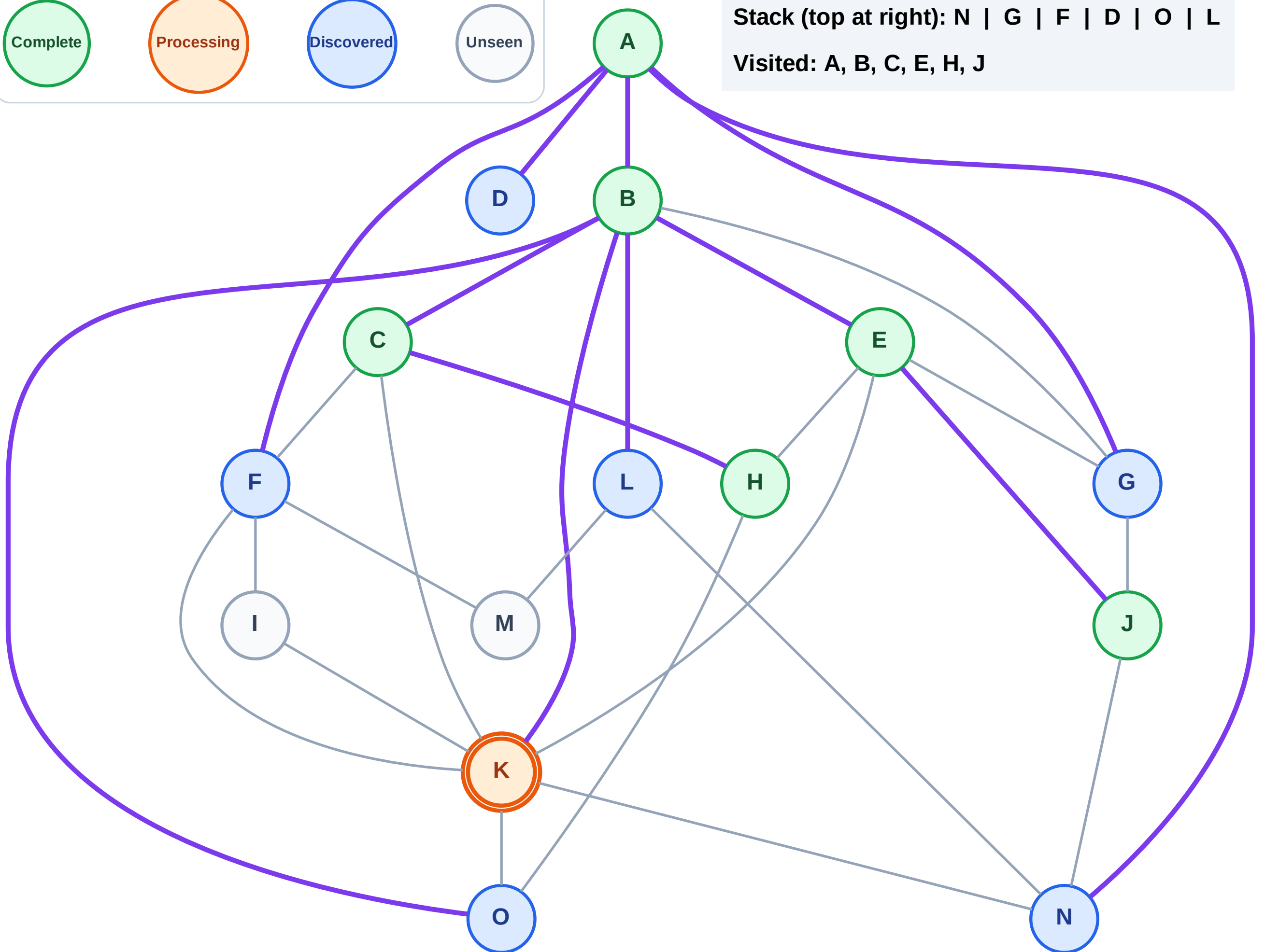
Step 14: Process node K

Legend



Stack (top at right): N | G | F | D | O | L

Visited: A, B, C, E, H, J



DFS Traversal

Step 15: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): N | G | F | D | O | L

Visited: A, B, C, E, H, J, K

